

CAMBRIDGE O LEVEL

Biology 5090

Chapters 7 & 13

Transport in flowering plants · Excretion

QUESTION BANK

cropped from QP · mark scheme under each question

QUESTION

1 · 5090/22/M/J/11 Q6

- 6 (a) (i) State what is meant by *transpiration*.
- (ii) Explain how the transpiration stream is important for the production of proteins in a
- (b) Describe and explain how tissues in the stem of a plant become stained when a plant is carefully removed from the soil and placed in a solution of dye for a few hours.
- 7 (a) Describe the functions of the kidneys

MARK SCHEME

5090/22/M/J/11 Q6

- 6 (a) (i) loss of water;
as vapour / evaporation;
from leaves / aerial parts of a plant / to the atmosphere;
through stomata (A between or through guard cells); [max. 3]
- (ii) brings nitrates;
dissolved in water;
from soil / to combine with carbohydrates; [max. 2]
- (b) dye diffuses (R ref. to osmosis);
through permeable (R semi- / partially- / selectively-);
(Ignore refs. to cell membrane)
(cell) wall;
of root (hair) cells;
xylem (any correct ref.);
ref. to travelling upwards (A refs. to transpiration pull / stream capillarity / root pressure AW); [max. 5]

[Total: 10]

QUESTION

2 · 5090/22/M/J/11 Q7

7 (a) Describe the functions of the kidneys.

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.....

MARK SCHEME

5090/22/M/J/11 Q7

- 7 (a) excretion (A excretory);
 removal of urea ;
 salts / ref. osmoregulation / varying amounts of water;
 toxins / poisons / hormones (R harmful);
 filtration of / removal from + blood / plasma;
 forming urine;

[max. 4]

- (b) ref dialysis;
 blood along a partially permeable tube;
 washing / bathing fluid (dialysis fluid could score 2 marks);
 ref controlled contents of bathing fluid (i.e. concentrations as required in blood)
 (Look for more than just a ref. to urea.);
 diffusion (any correct ref.);
 keeps blood at constant osmotic potential;
 washing fluid changed;
 temperature regulation;
 urea removed;
 blood from and to patient ;
 kidney transplants / donation;
 bubble trap / adding anticoagulants / counter flow / large surface area / pump ;
 (A coiled tube)

[max. 6]

[Total: 10]

QUESTION

3 · 5090/21/M/J/12 Q6

- 6 (a) State the function of phloem in a plant.
- (b) Describe how a molecule of water moves through a plant from the soil until it enters the atmosphere through the leaves.
- 7 (a) (i) State **one similarity** and **one difference** in the functions of the urethra in

MARK SCHEME

5090/21/M/J/12 Q6

- 6 (a) transport sucrose / sugar / amino acids (R glucose / food);
in solution;
ref. direction of movement (A around the plant);
- (b) osmosis;
through cell membrane;
of root hair;
ref. movement from cell to cell;
ref. movement through or between cell walls;
enters xylem;
transpiration pull / stream / capillarity / molecular cohesion / root pressure;
evaporation / ref. water vapour;
from mesophyll cells / into air spaces;
ref. water potential gradient;
diffusion;
through stomata;

[2]

[8]

[Total: 10]

QUESTION

4 · 5090/21/M/J/12 Q7

- 7 (a) (i) State **one similarity** and **one difference** in the functions of the urethra in male and in a female adult person.

similarity

.....

difference

MARK SCHEME

5090/21/M/J/12 Q7

- 7 (a) (i) carries urine (R just urea) in male and female;
carries semen / sperms / gametes AW in male; [2]
- (ii) ref. size comparison;
ref. qualified numbers comparison;
ref. mobility comparison; [3]
- (b) (*advantages*)
only needs to be undergone once AW (e.g. less trouble);
high reliability / effectiveness AW; [2]
- (*disadvantages*)
does not protect against sexually transmitted diseases;
ref. risks of surgery / anaesthetic;
not a temporary solution;
difficult / expensive to reverse;
need access to medical services AW; [4 max]

[Total: 10]

QUESTION

5 · 5090/22/M/J/12 Q3

- (b) List three structural changes that must occur in young, unmodified plant cells as they develop into xylem tissue.
- (c) A small, leafy branch is cut from a tree. After some hours, the stem of the branch remains firm but the leaves become limp. Suggest an explanation for this.

4 Fig 4.1 shows a section through a ripe fruit of a tomato plant

MARK SCHEME

5090/22/M/J/12 Q3

- 3 (a) (i) cell wall + correct drawing (outside existing line);
- (ii) nucleus + correct drawing (must be in cytoplasm);
- (iii) vacuole / cytoplasm + correct drawing (if vacuole, must be larger than the nucleus.); [3]

If no marks scored through unacceptable drawings, allow max. 1 if all 3 are shown in the correct positions – vacuole in the middle, cell wall on the outside and nucleus between.

- (b) elongation of cell (R cylindrical);

loss of cell contents / or one named content (A ref. death of cell / hollow)
(R chloroplasts / dissolving of cell contents.);

loss of end walls;

addition of thickening / strengthening material / lignin; [max. 3]

- (c) (*stem*) contains xylem;

(xylem is) thickened / strengthened / contains lignin;

provides support;

(*leaf*) xylem less important for support;

loss of water / transpiration;

water not replaced / doesn't reach leaves;

cells become flaccid / lose turgor AW (R plasmolysed);

cells unable to support weight of leaf / wilts; [max. 5]

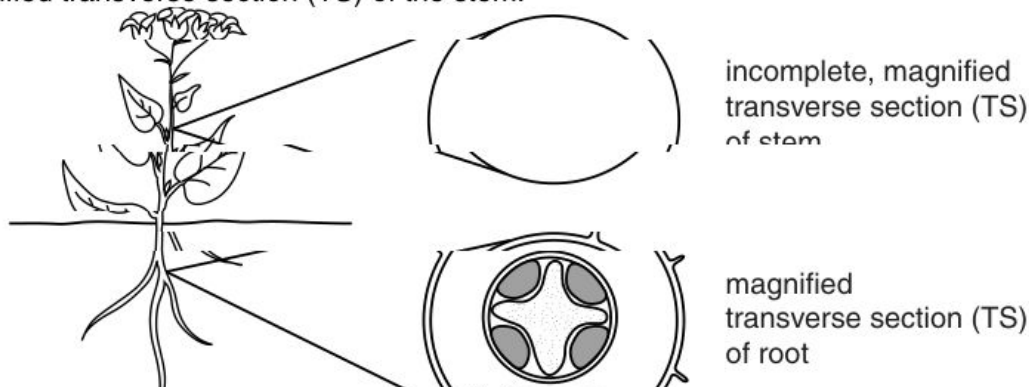
6 · 5090/22/M/J/12 Q9

5090/22/M/J/12 Q9

QUESTION

7 · 5090/21/M/J/16 Q1

- 1 Fig. 1.1 shows a plant with a magnified transverse section (TS) of the root and an incomplete, magnified transverse section (TS) of the stem.



- (a) The transverse section of the root in Fig. 1.1 shows the distribution of xylem and phloem
- Using a labelling line and the letter **X**, show, on Fig. 1.1, the location of the xylem tissue in the transverse section (TS) of the root. [1]
 - Complete the transverse section (TS) of the stem in Fig. 1.1 to show the distribution of xylem and phloem tissue. [2]
 - Using a label line and the letter **P**, show the location of the phloem tissue that you have [2]
- (b) State the functions of phloem tissue.

- 2 Dry mass is the mass of all chemicals, excluding water, present in an organism

MARK SCHEME

5090/21/M/J/16 Q1

1	(a) (i)	Label line X on xylem on transverse section of root ;	[1]
	(ii)	vascular bundle drawn towards outside of transverse section of stem ; vascular bundle divided into two sections ; three or more vascular bundles drawn ;	[max 2]
	(iii)	label line P drawn in correct location (outer side of vascular bundle) ;	[1]
	(b)	translocation ; transport ; dissolved / in solution ; sugar(s) / sucrose / products of photosynthesis ; amino acids ; reference from source + to sink / leaves to roots AW ;	[max 3]
			[Total: 7]

8 · 5090/21/M/J/17 Q2

-
- A diagram illustrating the movement of water in a plant. A plant with leaves is shown on the left. A glass of water, labeled 'water reservoir', is connected to the plant's stem by a tube. A bubble is shown moving upwards in the tube, indicated by an arrow and the text 'bubble moves this way as water is taken up by the plant'. The bubble is labeled 'bubble'.

- 3** Fig. 3.1 shows an organ **X** and its associated blood vessels **P**, **Q** and **R**.

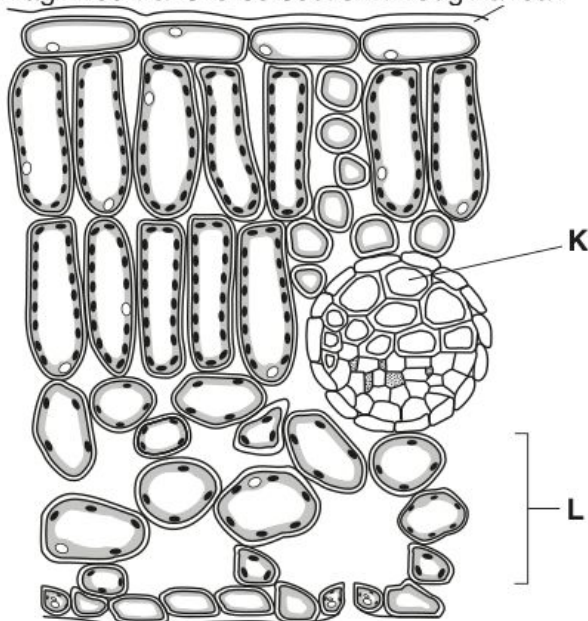
5090/21/M/I/17 02

2(a)	(A) petal ; (B) sepal ; (C) leaf / lamina / cuticle ; (D) vein / midrib / vascular (tissue) OR xylem + phloem ;	4	
2(b)	xylem ; to leaves / flower(s) ; between leaf cells ; surface of mesophyll cells ; (leaf) air spaces ; (through) stomata / guard cells ;	3	
2(c)(i)	6 ;	1	
2(c)(ii)	reduces / lower(s) AW ; less photosynthesis ; less water needed ; stomata / guard cells + open less / are closed ; less diffusion ; less evaporation / transpiration ;	3	
	Total:	11	

QUESTION

9 · 5090/21/M/J/18 Q6

6 The diagram shows a magnified transverse section through a leaf.



Name each of **J**, **K**, **L** and **M** and explain the importance of each in the process of transpiration.

[Total: 10]

/ (a) Define the term *allele*

MARK SCHEME

5090/21/M/J/18 Q6

6	(J) <u>cuticle</u> ; waterproof / waxy AW ; prevents / reduces + water loss / transpiration / evaporation ; (K) <u>xylem</u> ; transport + water ; from named location to named higher location ; (L) <u>spongy</u> ; air / intercellular + spaces ; water film AW ; evaporation / water vapour ; diffusion ; (M) stoma / stomata ; guard cell ; open / close ; reference to water <u>vapour</u> + exits ; diffusion ;	10	Note marking points must not be transferred between the sections R 'phloem' for xylem
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QUESTION

10 · 5090/21/M/J/20 Q3

- 3 The photograph shows an insect called a leafhopper. When it feeds, its mouthparts penetrate the xylem of the plant on which it lives.



Leafhoppers carry disease-causing bacteria from plant to plant.

- (a) (i) State the term used for a disease-causing organism.
 (ii) State the term used for an animal, such as an insect, that carries disease-causing organisms.
- (b) The bacteria carried by the leafhopper reproduce inside the plant stem and block the xylem.
 (i) Suggest and explain why this can cause the leaves of the plant to wilt.
 (ii) The leaves of plants affected by these bacteria may also develop white patches. Suggest reasons for this.
- (iii) Suggest an explanation for the fact that the fruits of plants affected by these bacteria are small and poorly developed.
- 4 (a) Explain the importance of the production of carbon dioxide in bread-making.

MARK SCHEME

5090/21/M/J/20 Q3

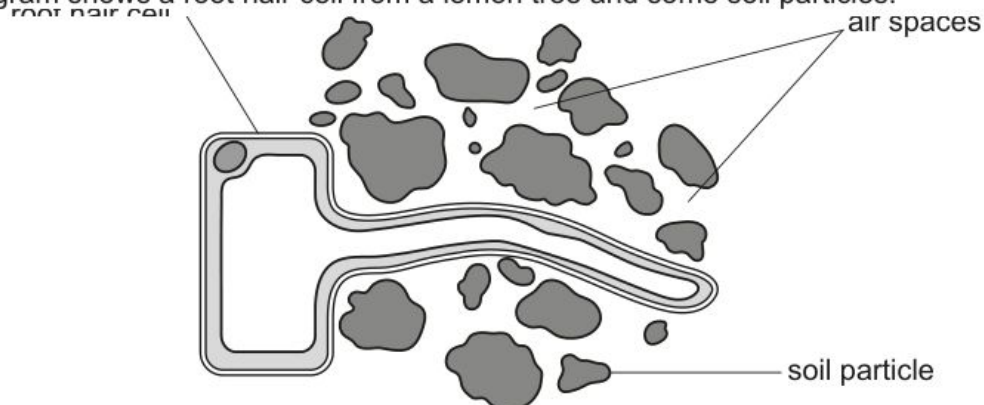
3(a)(i)	<u>pathogen</u> ;	1
3(a)(ii)	<u>vector</u> ;	1
3(b)(i)	water lost from leaves ; by evaporation / transpiration ; cannot be replaced / cannot pass up stem ; water lost from cells ; lose their turgor / turgidity ;	4
3(b)(ii)	can't make chlorophyll ; reference to magnesium / zinc not getting to leaves ;	2
3(b)(iii)	reduced photosynthesis ; less sugar / starch / protein made ; less water available to fruit ; fewer nutrients or named nutrient sent to fruit ; less growth or development (linked to protein) ;	4

Question	Answer	Marks
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QUESTION

11 · 5090/21/M/J/21 Q1

- 1 The diagram shows a root hair cell from a lemon tree and some soil particles.



- (a) (i) State **one** feature, visible in the diagram, which shows that a root hair cell is a plant cell.
 (ii) Describe how mineral ions pass from the soil into the root hair cell.
 (b) The lemon tree has grown in the same soil for many years so the soil now lacks magnesium. Describe and explain the effect on the lemon tree of a lack of magnesium.

- 2 A person visits an eye doctor to have an eye test. On the wall of the doctor's room is a diagram of

MARK SCHEME

5090/21/M/J/21 Q1

1(a)(i)	cell wall or large / central / sap + vacuole ;	1	
1(a)(ii)	<i>any four from:</i> dissolved in water ; through the cell wall ; <u>carriers</u> in membrane ; through the cell / partially permeable AW + membrane ; by active transport / diffusion ; correct reference to concentration gradient ; using energy (for active transport) ; from (aerobic) respiration (for active transport) ;	4	
1(b)	yellowing / chlorosis / reduced growth / cannot photosynthesise / make glucose / starch / sugar / carbohydrate ; deficiency of chlorophyll ;	2	

QUESTION

12 · 5090/21/M/J/21 Q7

- 7 (a) Excess amino acids are broken down in liver cells to form molecules of urea. Describe the pathway taken by a molecule of urea, from the liver of a mammal until it reaches the soil. [7]
- (b) Urea is a molecule which contains nitrogen. Suggest why it is important that molecules that contain nitrogen are added to the soil.

Section C

MARK SCHEME

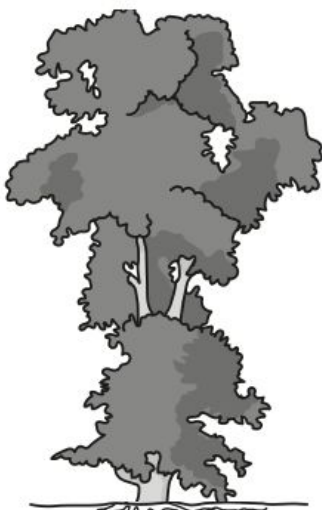
5090/21/M/J/21 Q7

7(a)	any seven from: transported in blood / plasma ; dissolved ; hepatic vein to vena cava / heart ; renal artery (to kidney) ; to kidney ; urea removed / filtered + from blood ; part of urine ; ureter (to bladder) ; bladder ; (bladder to) <u>urethra</u> ;	7	
7(b)	any three from: cycling / replenishing nitrogen ; used / converted by nitrifying bacteria / nitrogen fixing bacteria ; form + nitrates ; nitrates + taken up by plants ; production of proteins / amino acids ; <u>increased</u> soil fertility (for crops) / plants grow faster / bigger / AW ; animals / herbivores obtain protein / nitrogen from plants ;	3	

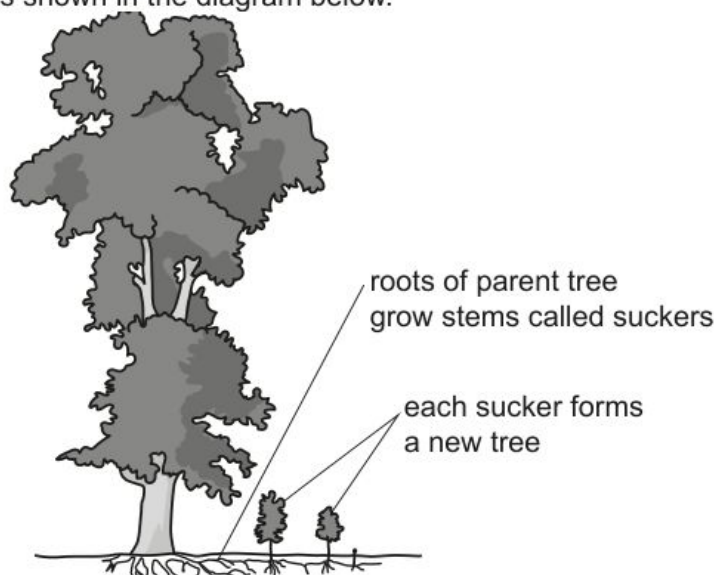
QUESTION

13 · 5090/22/M/J/21 Q3

3 The diagram shows an elm tree.



- (a) Elm trees may be infected by a fungus that causes a disease. State **two** characteristics of a fungus.
- (b) The fungus moves through an infected tree inside the xylem vessels. The tree responds to stop the movement of the fungus by producing substances that block the xylem vessels. This causes the leaves of the tree to wilt and to become yellow. Explain how the response of the tree to infection leads to these symptoms of disease.
- (c) An insect called the elm bark beetle is the vector of this disease. Describe what is meant by the term **vector**.
- (d) Elm trees can reproduce when the roots from one parent tree spread out near the surface of the soil. This is shown in the diagram below.



- (i) Name the type of reproduction shown in the diagram.
- (ii) Suggest why this type of reproduction makes it difficult to control the spread of the disease caused by the fungus.

4 Cardiopulmonary resuscitation (CPR) is a first aid procedure.

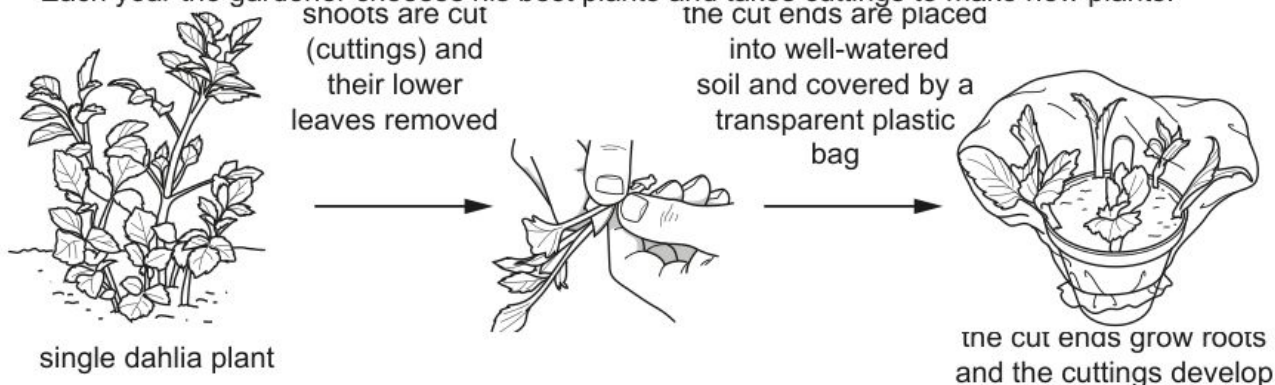
3001	1. Tophen / myofibrils 2. eukaryotic / eukaryotic / cells on dead matter / AM 3. saprophytic 4. cell wall / cellulose 5. more than one nucleus / multinucleate	2	
3002	1. less / not / water transported / water absorbed / transpiration 2. water loss greater than water uptake 3. less / no water in xylem 4. wilting / loss of turgor / plasmolysed 5. megastoma / WU 6. less / no / chlorophyll	5	
3003	1. transfer / copies / spreads / AM / pathogen / fungus / parasite 2. one organism AM to another 3. not host / affected / infected	2	
3004	1. identical	1	
3005	1. clone together 2. identical AM 3. genetically identical / all genetically susceptible / clones	3	
Question	Answer	Marks	Guidance

QUESTION

14 · 5090/21/M/J/22 Q3

3 A gardener grows a species of dahlia, a flowering plant.

Each year the gardener chooses his best plants and takes cuttings to make new plants.



(a) (i) Removing the lower leaves and placing the cuttings in a transparent bag helps prevent them wilting until the roots develop. Suggest and explain how these two steps help to prevent wilting.

removal of lower leaves

covering with transparent bag

(ii) Taking cuttings is a commercial application of asexual reproduction. Define asexual reproduction.

(b) The stems of the dahlia cuttings contain xylem vessels. Describe **two** functions of xylem vessels in young stems of dahlias.

For each function explain **one** way in which xylem structure is adapted for this function.

adaptation

adaptation

(c) The gardener's dahlia cuttings grow into full-sized plants with flowers.

The gardener decides that he would like to produce a plant that combines the disease-resistant characteristics of one of these plants with the red petal colour of another. With reference to named parts of the flowers, describe how he could achieve this.

4 The human liver contains a high concentration of enzymes called transaminases.

MARK SCHEME

5090/21/M/J/22 Q3

3(a)(i)	removal of leaves: reduction + surface area / stomata ;	1	
	covering with transparent bag: increase humidity ;	1	
	to reduce transpiration OR evaporation / water loss ;	1	
3(a)(ii)	any two from: production of <u>genetically identical</u> offspring / <u>clones</u> ; mitosis ; from one parent/ AW ;	2	

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3(b)	function: xylem vessels transport water and/or minerals ;	1	
	adaptation: long / thin / no end cell walls / hollow tubes / no cytoplasm ;	1	
	function: (mechanical) support / keeps stem upright ;	1	
	adaptation: lignified OR woody / strengthened / hard + walls ;	1	
3(c)	any five from: 1 artificial selection / selective breeding ; 2 transfer pollen / pollination ; 3 from anther / stamen of one plant + e.g. red petal colour ; 4 to stigma / carpel / pistil of another ; 5 fertilisation OR gametes / genetic material + combined AW ; 6 grow seeds ; 7 select the ones with best combination of characteristics ;	5	

Question	Answer	Marks	Guidance
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QUESTION

15 · 5090/22/M/J/22 Q5

- 5 (a) Biological processes in plants are affected by environmental factors. The graphs each show the effect of changing one environmental factor on a biological process. The labels on the axes for each graph have been replaced with the letters P, Q, R and S.

Identify the correct axis label for each of P, Q, R and S. Select from the following possible labels.

temperature carbon dioxide concentration rate of transpiration
light intensity rate of photosynthesis air humidity

- (b) Describe the process of transpiration in the leaf of a plant.

SECTION B

MARK SCHEME

5090/22/M/J/22 Q5

5(a)	(P) rate of photosynthesis ; (Q) temperature ; (R) rate of transpiration ; (S) air humidity ;	4	
5(b)	1 <u>evaporation</u> ; 2 surface of / cell wall / film of water / layer of water + mesophyll cells ; 3 <u>water vapour</u> ; 4 air spaces / intercellular spaces ; 5 diffusion / high to low concentration / down concentration gradient ; 6 water + through <u>stomata</u> ;	3	

Question	Answer	Marks	Guidance
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QUESTION

16 · 5090/22/M/J/22 Q9

9 (a) Compare the ureter and the urethra in terms of their functions in adult humans.

MARK SCHEME

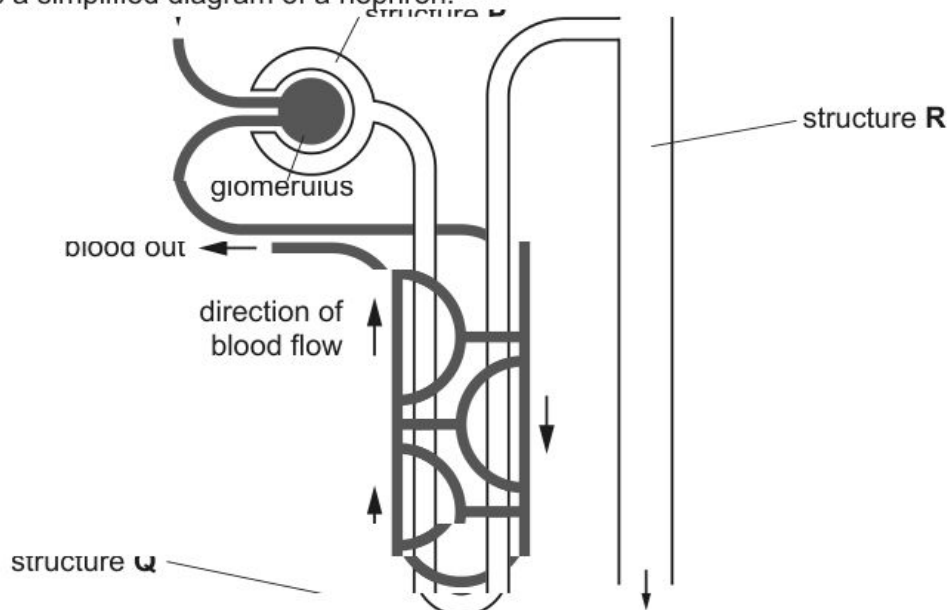
5090/22/M/J/22 Q9

9(a)	1 both carry + urine / urea ; 2 ureter <u>from</u> kidney ; 3 ureter <u>to</u> bladder ; 4 urethra <u>from</u> bladder 5 urethra <u>out of</u> body ; (any two from) 6 male / penis / testis + urethra carries + semen / sperm ;	4	
9(b)	1 selectively / partially / semi + permeable ; 2 large surface area ; 3 fluid / dialysate ; 4 counter-current / continuously changed ; 5 urea / toxins / poisons / harmful substances + from blood / to fluid ; 6 water / salts / ions + between blood and fluid ; 7 diffusion / osmosis ; 8 high to low concentration / <u>down</u> concentration gradient ; 9 large <u>molecules</u> / <u>proteins</u> + stay in blood / do not enter fluid ;	6	

QUESTION

17 · 5090/22/M/J/24 Q4

4 Fig. 4.1 is a simplified diagram of a nephron.



- (a) (i) Identify each of the structures labelled **P**, **Q** and **R** in Fig. 4.1.
 (ii) Name the organ which contains many nephrons.
 (iii) Name the organ that stores fluid **S** before it is released from the body.

(b) Table 4.1 shows the composition of a sample of fluid **S** from a healthy person. The person that provided the sample of fluid **S** eats a balanced diet.

component	concentration / arbitrary units
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- (i) State why it is important that urea is removed from the blood in fluid **S**.
 (ii) Use your knowledge of the function of a nephron to explain the concentration of each of the following components of fluid **S**.
 (iii) Explain what would happen to the concentration of ions in fluid **S** if the person drank a large volume of water several hours before the sample was collected.

Total: 14

5 Fig. 5.1 is a diagram of a pineapple growing on a pineapple plant.

MARK SCHEME

5090/22/M/J/24 Q4

4(a)(i)	(P) Bowman's capsule ; (Q) loop of Henle ; (R) collecting duct ;	3	
4(a)(ii)	kidney ;	1	
4(a)(iii)	(urinary) bladder ;	1	
4(b)(i)	toxic / toxin ;	1	
4(b)(ii)	glucose 1 enters Bowman's capsule / enters nephron / removed from blood AW ; 2 by filtration / filtered; 3 back into blood / reabsorbed + into blood ; 4 AVP ; protein 5 doesn't enter Bowman's capsule / doesn't enter nephron / no filtration / not removed from blood AW ; 6 protein + large AW ; max 5	5	
4(b)(iii)	ion concentration + decrease ; more water / higher volume of water / excess water + in fluid S / urine / excreted; homeostasis / to keep concentration of water in blood constant AW;	3	

QUESTION

18 · 5090/22/M/J/24 Q6

- 6 Flowering plants are sometimes grown in soil to provide cut flowers that can then be displayed.
- (a) A gardener watered flowering plants with a fertiliser solution containing dissolved chemicals. Table 6.1 shows the information printed on the label of the container of fertiliser powder used to prepare this fertiliser solution.

Table 6.1	
chemical component	mass/g per 100 g fertiliser powder
nitrates	4.1
phosphates	4.0
magnesium	0.5

- (i) To prepare the fertiliser solution, the gardener dissolved 20 g of the fertiliser powder in 10 dm³ of water. Calculate the total mass of sulfur in the fertiliser solution prepared by the gardener.
Space for working.
- (ii) Explain which chemical component of the fertiliser solution is the most important to making the leaves of the plants green in colour.
the production of enzymes by cells in the leaves of the plants.
- (iii) The chemical components of the fertiliser solution are absorbed by the root hair cells of the plants and are then transported to the leaves.
Complete the words below to name the three main tissues in the pathway taken by the chemical components of the fertiliser solution from the root hair cells to the leaves.
The first letter of each word has been completed for you.
- (b) When flowers from the plants are cut and used for display, the leaves are removed before each stem is placed in water.
A solution containing sucrose is often added to this water.
Sucrose is **not** needed in fertiliser solution used to water plants growing in soil.
Explain the benefit of providing sucrose to cut flowers.

- 7 Plastics are produced in very large amounts by industrial processes around the world.

MARK SCHEME

5090/22/M/J/24 Q6

6(a)(i)	0.2 ; g ;	2	
6(a)(ii)	magnesium ; chlorophyll (makes leaves green); nitrates ; (enzymes made from) amino acids / proteins ;	4	
6(a)(iii)	(c)ortex ; (x)ylem ; (m)esophyll ;	3	
6(b)	1 no + photosynthesis ; 2 to produce glucose / glucose not produced ; 3 to produce sucrose / sucrose not produced ; 4 sucrose travels in phloem ; 5 for respiration / energy ; 6 flowers / plants + last longer / for days AW ; max 3	3	

QUESTION

19 · 5090/22/M/J/25 Q7

7 Photosynthesis and transpiration are two processes that take place in a plant leaf.

(a) Explain how the structure of a leaf is adapted for photosynthesis.

(b) Transpiration is both necessary for plants and a problem for plants. Discuss this statement.

MARK SCHEME

5090/22/M/J/25 Q7

7(a)	1 large surface area + maximum light absorption ; 2 stomata + gaseous exchange / carbon dioxide in / oxygen out ; 3 thin + diffusion of gases / carbon dioxide in / oxygen out ; 4 air spaces / spongy mesophyll + diffusion of gases / gaseous exchange ; 5 chloroplasts / chlorophyll + absorb light ; 6 chloroplasts / chlorophyll + most dense in palisade (mesophyll) ; 7 palisade in upper leaf + receive most light ; 8 xylem (vessels) + bring water (from soil) ; 9 epidermal cells no chloroplasts / transparent + light reaches cells ; max 6	6	
7(b)	necessary (max 2): 1 provides (leaves with) water for photosynthesis ; 2 transports minerals / ions from roots / to leaves ; 3 provides plant cells with support / turgidity / turgor ; 4 cools the leaf / plant ; problem (max 2): 5 too much water is lost (through stomata / leaf) ; 6 (water loss) leads to wilting / death ; 7 plants may close stomata to prevent transpiration, which slows down photosynthesis ; max 4	4	

QUESTION

20 · 5090/21/O/N/11 Q8

8 (a) State the chemical substances that are excreted by humans.

(b) Explain how each of these substances is excreted.

9 (a) State the chemical substances released by a plant to the environment

MARK SCHEME

5090/21/O/N/11 Q8

8 (a) urea;
carbon dioxide;
water;
salts;
toxins/broken-down hormones;
bile salts/pigments;

[max 3]

(b) urea/water/salts/toxins/broken down hormones + kidneys;
blood/blood vessels/named vessel/capillaries;
bladder + urine/urination;
ureter + urethra (both correctly spelt);
water/CO₂ + lungs/alveoli;
diffusion + from capillaries (for CO₂);
breathing (out);
water/salts/urea + sweat;
sweat glands;
from blood/capillaries;
sweat ducts/pores;
ref. faeces ONLY in an explanation of how bile salts/pigments are removed;

[max 7]

[Total: 10]

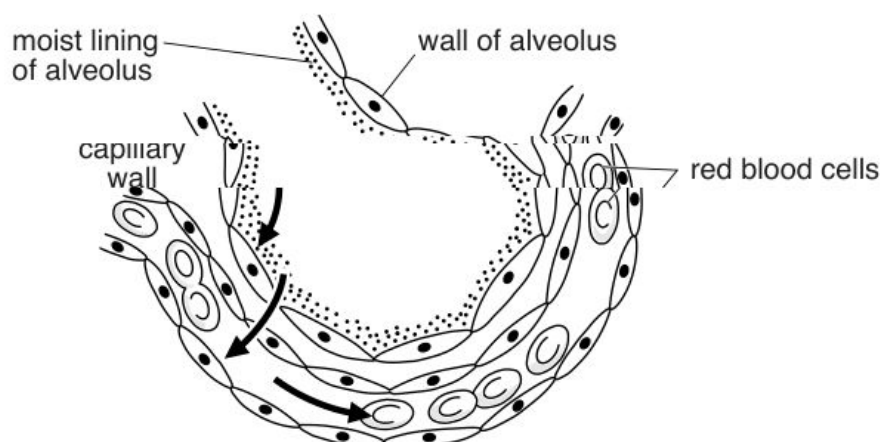
QUESTION

21 · 5090/21/O/N/12 Q1

- 1 The urinary systems of a dog and of a human being are similar in structure and function. Fig. 1.1 shows the urinary system of a dog.



- (a) (i) Name the parts **A**, **B** and **C** in Fig. 1.1.
 (ii) State the function of **D**.
- (b) Fig. 1.2 shows the relationship between an alveolus and a blood capillary in the lung of a mammal. The arrows show the path taken by oxygen from inhaled air entering the blood during gas exchange.



- (i) Draw **two** arrows on Fig. 1.2 to show the path taken by most carbon dioxide leaving the blood.
 (ii) Explain why this movement of carbon dioxide is considered to be part of excretion.
- (c) Hormones are also substances that need to be excreted from the body after they have carried out their functions.
 Explain how this is done.

- 2 (a) The graphs in Fig. 2.1 show the effect of changing pH on the rate of reactions carried out by enzymes.

MARK SCHEME

5090/21/O/N/12 Q1

- 1 (a) (i) **A** kidney;
B ureter;
C urethra;
- (ii) **D** storage AW + urine; [1]
- (b) (i) arrow shown leaving plasma;
 arrow from moisture to alveolar air;
 One arrow covering the entire journey = 1 mark [2]
- (ii) removal + from body / organism;
 waste;
 of respiration / metabolism; [3]
- (c) destroyed / broken down **AW**;
 by liver;
products removed / reabsorbed; [max 2]

[Total: 11]

QUESTION

22 · 5090/21/O/N/12 Q6

- 6 Aphids are insects that feed by piercing the phloem of soft, young shoots. Aphids then use their sharp, hollow mouthparts for sucking up their nutrition from the plant.
- (a) Explain why phloem is more suitable for this purpose than any other plant tissue.
- (b) Suggest and explain the effect on the plant shoot of a large number of aphids feeding at the same time over a period of several weeks.
- 7 (a) Explain the differences between *mitosis* and *meiosis*. Your explanation should

MARK SCHEME

5090/21/O/N/12 Q6

- 6 (a) within reach;
contains sucrose / sugar / carbohydrate;
contains amino acids;
in solution / water;
not lignified / is softer **AW**;
R starch / glucose / mineral ions
lg nutrients

[3]

- (b) impaired growth **AW** / death;
amino acids used to make protein;
less / no new cytoplasm;
less sucrose or carbohydrate;
less glucose;
for use in respiration;
energy used;
for any named purpose;
may introduce viruses / disease;
damage to plant **AW**;
lg wilting

[max 7]

[Total: 10]

23 · 5090/22/O/N/12 Q7

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..[3]

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[Max 3]

[Max 3]

[Max 3]

[Max 3]

[Max 7]

Total: 10]

QUESTION

24 · 5090/21/O/N/13 Q6

- 6 (a) Describe the functions of xylem in a plant.
- (b) Explain how starch and protein stored in the roots of a plant during the day can be used in the growth of a shoot during the night.
- 7 (a) Explain the importance of active transport in plants and in humans

MARK SCHEME

5090/21/O/N/13 Q6

6 (a)	support / rigidity; lignin; conduct/transport of water; ions / minerals / salts; ref to narrow/hollow/continuous / lumen; <u>from roots(hair)</u> + ref to any aerial part of plant;	[max 3]	R transport of amino acids / sugars
(b)	ref enzyme / named enzyme; action of enzyme ; <u>named</u> product of enzyme action; ref to solubility of product <u>diffusion</u> (from storage cells to phloem or vice –versa); to <u>phloem</u> ; sucrose/amino acids (moved up phloem) ; ref translocation/transport up stem; no photosynthesis at night ; (energy) released during <u>respiration</u> ; sugar provides energy for growth; amino acids make proteins; to make <u>new</u> cytoplasm / cells;	[max 7]	A amino acids / glucose / sugar R produced
		[Total 10]	

QUESTION

25 · 5090/22/O/N/13 Q7

- 7 (a) Explain why the lungs are considered to be organs of excretion.

.....

.....

.....

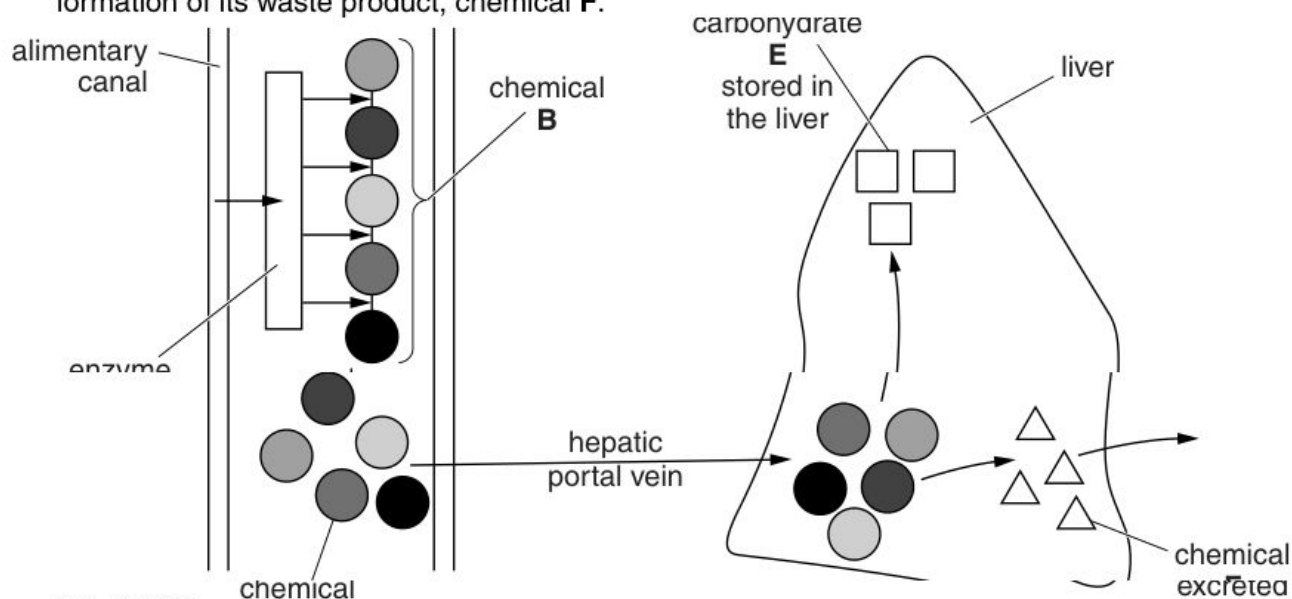
.....



QUESTION

26 · 5090/21/O/N/14 Q2

- 2 Fig. 2.1 shows the digestion, absorption and assimilation of a chemical (B). It also shows the formation of its waste product, chemical F.



- (a) chemical B
 enzyme C
 chemical D
 carbohydrate E
 chemical F [5]
- (b) (i) Describe what happens to carbohydrate E before it can be used in metabolic reactions in body cells. [2]
- (ii) Name the process in which the product of carbohydrate E is used within a body cell.
- (iii) Describe how different types of body cells and tissues make use of this process.
- 3 (a) At puberty, hormones are produced that are responsible for the development of secondary

MARK SCHEME

5090/21/O/N/14 Q2

2 (a)	B = protein / (poly)peptide ; C = protease or named* ; D = amino acids* ; E = <u>glycogen</u> ; F = urea ;	[5]	*A ecf for incorrect substrate *A ecf for incorrect substrate R urine
(b) (i)	broken down / converted / changed ; (role of) glucagon / adrenaline ; to <u>glucose</u> ; made soluble ;	[max. 2]	
(ii)	<u>respiration</u> ;	[1]	

QUESTION

27 · 5090/21/O/N/14 Q7

7 Describe the similarities and differences in structure and function of root hairs and villi.

(a) similarities

(b) differences

SECTION C

MARK SCHEME

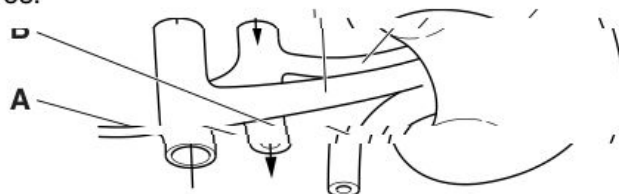
5090/21/O/N/14 Q7

7 (a)	<i>structural similarities:</i> long / elongated ; providing large surface area ; <i>functional similarities:</i> absorption / uptake ; active transport / diffusion ; ions / salts / minerals / named ; water ;	[max. 4]	
(b)	<i>structural differences:</i> cell wall / no cell wall ; (root hair) part of one cell ; (villi) many cells / multicellular ; ref. absence of blood vessels / lacteals / ORA ; AVP ; <i>functional differences:</i> root hairs + absorb from the soil ; villi + absorb from the gut ; villi + absorb amino acids ; villi + absorb glucose ; villi + absorb lipids / glycerol / fatty acids ;	[max. 6]	
		[Total: 10]	

QUESTION

28 · 5090/22/O/N/14 Q1

- 1 Fig. 1.1 shows a kidney and its associated structures. The arrows show the direction of flow of fluids in these structures.



- (a) (i) Name the structure labelled **A** in Fig. 1.1. [1]
 (ii) Name the chamber of the heart through which blood in structure **A** last passed. [1]
- (b) Table 1.1 shows the relative concentrations of various substances in structures **B** and **C**. Complete the table to show the possible concentrations of these substances in structure **D**.

substance	relative concentration in structure		
	B	C	D
amino acids	0.00	0.05	
glucose	0.00	0.10	
mineral ions	1.50	0.72	
proteins	0.00	8.00	
urea	2.00	0.03	

- (c) Explain how the relative concentrations of glucose might change in structures **B**, **C** and **D** in a person with diabetes.

- 2 Fig. 2.1 shows parts of the alimentary canal that lie in the upper part of the human body

MARK SCHEME

5090/22/O/N/14 Q1

1	(a) (i)	(dorsal) <u>aorta</u> ;	[1]	
	(ii)	<u>left ventricle</u> ;	[1]	
	(b)	(amino acids) 0.05 ; (glucose) 0.10 to 0.15 ; (mineral ions) 0.72 to 2.22 ; (proteins) 8.00 ; (urea) 0.03 to 2.03 ;	[5]	
	(c)	B would contain some / more / high (glucose) / C would contain more / high (glucose) / D would contain more / high (glucose) ; lack of Insulin ; glucose would not be converted into glycogen ; kidney unable to / doesn't reabsorb all glucose ;	[max. 3]	
			[Total: 10]	

QUESTION

29 · 5090/21/O/N/15 Q5

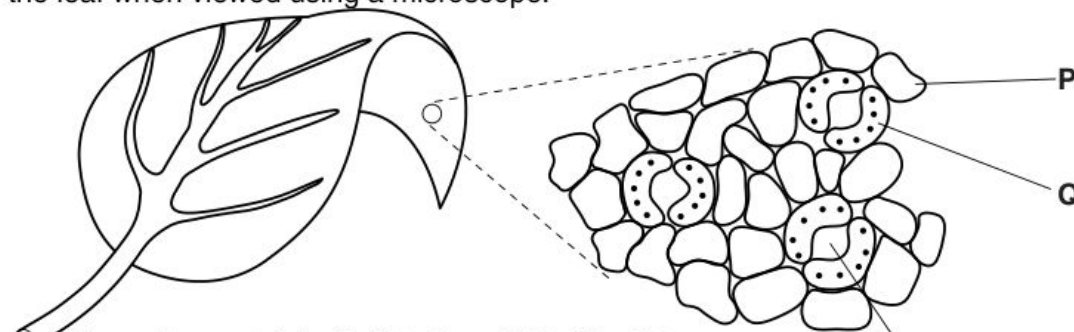
- 5 Fig. 5.1 shows a plant growing in an area of well-watered soil.



- (a) The plant was held by the stem and pulled from the ground before being re-planted in another area of well-watered soil. It was observed that the plant wilted for several days after being re-planted and then recovered its original appearance. Suggest an explanation for this observation.

- (b) Photosynthesis takes place in the leaves of the plant. Complete the equation below for photosynthesis using either words or symbols.

- (c) Fig. 5.2 shows one leaf taken from the plant and the appearance of part of the lower side of the leaf when viewed using a microscope.



- (i) Name the parts labelled P, Q and R in Fig. 5.2.
- (ii) Suggest how the upper surface of the same leaf would appear different from the lower surface shown in Fig. 5.2 when viewed using a microscope. Explain the reason for this difference.

SECTION B

MARK SCHEME

5090/21/O/N/15 Q5

5 (a)	<i>wilting:</i> root hairs removed / roots damaged ; stem / xylem damaged ; less water absorbed ; plant loses water / transpires ; cells lose water / cells become flaccid ; <i>recovery:</i> root (hairs) regrow ; (more) water absorbed ; cells gain water again / become turgid ;	[max. 4]	R no water absorbed allow max.3 marks for <i>wilting</i> answers
(b)	carbon dioxide + water → oxygen ;	[1]	accept words or chemical formulae or a correctly balanced symbol equation 1 light / chlorophyll
(c) (i)	P = (lower) epidermal cell / epidermis / epidermal layer ; Q = guard cell ; R = stoma / stomata ;	[3]	
(ii)	fewer / no stomata / guard cells ; reduce / stop water loss / transpiration ; (appears) shiny / bright / ref. to cuticle ;	[max. 2]	
		[Total: 10]	

QUESTION

30 · 5090/21/O/N/15 Q9

- 9 (a) Define the term *excretion*.
- (b) Explain how a kidney machine carries out the functions of a kidney for a person with kidney

Total: 10

MARK SCHEME

5090/21/O/N/15 Q9

9 (a)	removal from organism/body ; toxic/poisonous ; waste (products) ; from metabolism ;	[max. 3]	
(b)	1 <u>blood</u> from patient to machine/returned to patient ; 2 using <u>partially permeable</u> membrane/tubes AW ; 3 ref. dialysis fluid ; 4 ref. relative concentrations AW ; 5 filters/removes substances from + blood/unwanted out/wanted in ; 6 by <u>diffusion</u> ; 7 urea/nitrogenous products/toxins/poisons ; 8 salt(s)/ions/small molecules ; 9 excess water/ref. osmoregulation ; 10 large molecules/proteins stay in blood ; 11 practical aspect e.g. bubble trap/repetition/frequency/duration of procedure/dialysis fluid frequently changed/temperature ref/counter flow ;	[max. 7]	
		[Total: 10]	

QUESTION

31 · 5090/22/O/N/15 Q7

- 7 (a) (i) Explain how wilting occurs in a plant.
- (ii) Explain why mature plants do not wilt as much as newly-transplanted seedlings, even when both are exposed to similar conditions.
- (b) Suggest an explanation for what would happen to a plant if it is accidentally 'watered' with a concentrated salt solution.

SECTION C

MARK SCHEME

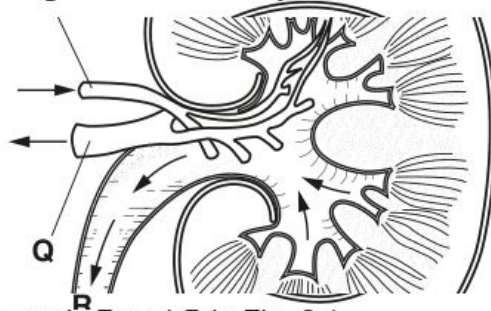
5090/22/O/N/15 Q7

7 (a) (i)	water lost/plant dehydrated ; by transpiration/evaporation/through stomata ; faster than it is gained ; cells/tissues lose turgor/become flaccid/plasmolysed AW ; no longer support plant ; any condition leading to wilting ;	[max 4]	A lack of water, low humidity, high light intensity, high temperature, wind
(ii)	(mature plant) root system undamaged ORA ; ref. root hairs ; water uptake sufficient for plant ORA ; plant cells/tissues turgid ORA ; xylem supports older plants/ref. lignin ;	[max 3]	I ref. more/less water uptake
(b)	plant/leaves will wilt ; water lost ; from cells/tissues ; by osmosis/diffusion ; lower water potential in soil water/ref. correct water potential gradient AW ;	[max 3]	
		[Total: 10]	

QUESTION

32 · 5090/21/O/N/17 Q2

2 Fig. 2.1 shows a section through a human kidney and its associated structures.



- (a) (i) Name the blood vessels **P** and **Q** in Fig. 2.1.
 (ii) Name the fluid **R** in Fig. 2.1 and describe the route taken by this fluid from leaving the kidney to being removed from the body.
 name of fluid
 route taken by fluid

(b) A person with kidney damage requires the process of dialysis to take place several times

MARK SCHEME

5090/21/O/N/17 Q2

2(a)(i)	P renal artery ; Q renal vein ;	2	
2(a)(ii)	urine ; ureter ; bladder ; urethra ;	4	R urea
2(b)(i)	reference to artery + vein ;	1	
2(b)(ii)	lower pressure ; less / no urea ; reference to increase / decrease + concentration of ions AW ;	1	
2(c)	(artificial membrane) partially permeable AW ; small molecules or named + pass through (membrane) ; correct reference to diffusion / osmosis ; proteins do not pass through (membrane) ; (dialysis fluid) no / low urea OR removes urea ; reference to concentration gradient ; same glucose concentration + as normal blood ; same ion AW concentration + as normal blood ;	5	

QUESTION

33 · 5090/21/O/N/19 Q8

- 8 (a) Describe and explain how a protein molecule that is eaten becomes molecules of urea that are excreted.
- (b) Explain why carbon dioxide does not normally pass out of the leaves of a plant during the day.
- 9 (a) Describe and explain how a nitrate ion in the soil becomes part of a stored molecule in the

MARK SCHEME

5090/21/O/N/19 Q8

8(a)	digested / broken down / hydrolysis ; into amino acids ; by protease or named enzyme ; absorbed in small intestine, duodenum / ileum ; in villi / by capillaries / blood ; hepatic portal vein ; to liver ; deamination or described ; urea + in renal artery ; urea + taken to kidney ; blood filtered / urea leaves blood ; forms urine ; bladder / urethra ;	8	
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8(b)	photosynthesis ; uses up CO ₂ ; faster than is made during respiration ;	2	
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Question	Answer	Marks	Guidance
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QUESTION

34 · 5090/22/O/N/20 Q7

- 7 The kidneys are organs of excretion.
- (a) Define the term *excretion*.

MARK SCHEME

5090/22/O/N/20 Q7

7(a)	1 removal AW ; 2 toxins / toxic ; 3 waste products AW / urea / carbon dioxide ;	3	3 R egestion
7(b)(i)	1 multiple tubes / multiple sheets / coiled / folded / long ; 2 large AW + surface area ; 3 selectively / partially / semi + permeable ; 4 diffusion / osmosis ; 5 small / waste products AW / urea / ions / water / toxins + do pass AW ; 6 large / protein / blood cells + do not pass AW ;	5	

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7(b)(ii)	1 maintain gradient AW ; 2 prevent build up / ensure removal + waste products AW / urea / toxins ; 3 continued / more / maximum + diffusion / movement / exchange ;	2	
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Question	Answer	Marks	Guidance
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QUESTION

35 · 5090/21/O/N/21 Q3

- 3 A man is cooking in a hot kitchen and sweat is being released by sweat glands in his skin.
- (a) (i) Explain the term homeostasis with reference to sweating.
- (ii) Explain how receptors and blood vessels in the man's skin are also involved in homeostasis when he is in the hot kitchen.
- (b) Sweat contains very small quantities of the waste product urea. State the name of the organ where most urea is made in the body.
- (c) The table shows the concentrations of some chemicals found in the sweat, urine and blood plasma of a healthy human.

chemical	concentration of chemical /mmol per dm ³		
	sweat	urine	blood plasma
urea	—	—	—
sodium	66	110	141
chloride	59	103	99

- (i) Use the data in the table to compare the chemical composition of urine and blood

MARK SCHEME

5090/21/O/N/21 Q3

3(a)(i)	<i>any two from:</i> (homeostasis is) the maintenance of a <u>constant</u> ; <u>internal environment</u> / temperature ; return to norm / return to set-point / negative feedback ; <u>evaporation</u> of sweat removes heat / thermal energy / cools / decreases temperature ;	2	
3(a)(ii)	<i>any four from:</i> receptors + detection of temperature ; <u>impulse</u> to brain / CNS ; <u>vasodilation</u> ; arterioles / blood vessels widen ; increase in blood + near skin surface ; greater + loss of heat / thermal energy ; by radiation ;	4	
3(b)	<u>liver</u> ;	1	

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3(c)(i)	<i>any four from:</i> less urea in blood plasma ; more sodium in blood plasma ; less chloride in blood plasma ; comparative data quote with units ; data manipulation, e.g. 50 × higher urea in urine than blood plasma ;	4	
3(c)(ii)	high(er) / over 22 (in patient) ; reduced filtration / reduced removal of urea (from kidney) ; reference to blood <u>plasma</u> ;	3	

Question	Answer	Marks	Guidance
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QUESTION

36 · 5090/21/O/N/22 Q1

- 1 The diagram shows parts of the human body involved in excretion.



- (a) Name the liquid that passes out of the human body through tube D and the waste product it carries.

- (b) Describe how parts A, B and C are involved in removing the waste product that passes out of the body.

- 2 Grapes are fruits that grow on a plant called a grape vine.

MARK SCHEME

5090/21/O/N/22 Q1

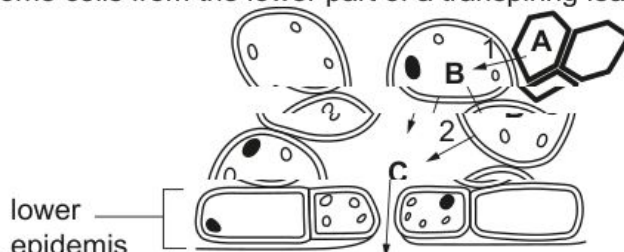
1(a)	urine ;	1	
	urea ;	1	A ammonia / excess water / excess minerals
1(b)	renal artery / A + brings urea / waste product to B / kidney AW ;	1	
	kidney / B + filters blood to remove urea / waste products ;	1	
	bladder / C + stores / keeps / collects urine ;	1	

Question	Answer	Marks	Guidance
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QUESTION

37 · 5090/21/O/N/22 Q6

- 6 The diagram shows some cells from the lower part of a transpiring leaf.



- (a) (i) Identify cell type B.
(ii) Water travels from A to D. The arrows on the diagram represent the movement of water molecules during transpiration. With reference to these arrows, name and define each process taking place.

- (b) Leaves from different plant species transpire at different rates in the same environmental conditions. Suggest and explain structural differences in the lower surface of these leaves which might produce differences in the transpiration rate.

- 7 Humans have receptors to detect stimuli in their environment and a nervous system to coordinate

Transpiration Process 1. The movement of water molecules from an area of higher water potential to an area of lower water potential across a partially permeable membrane. 2. The change from liquid (water) to a gaseous state (vapour). 3. The final movement of molecules from a high to a low concentration - down a concentration gradient.	1 A correct definition of liquid to a gaseous state and across a partially permeable membrane.
Structure 1. The surface area of the leaf. 2. The greater the surface area of the leaf, the greater the surface for evaporation. 3. The thickness of the leaf. 4. The thicker the leaf, the greater the barrier for water molecules. 5. The number of stomata. 6. The density of stomata increases, more places for water to escape from. 7. The size of stomata. 8. The size of stomata increases, so there are more places for water to escape from. 9. The number of stomata. 10. The number of stomata increases, so there are more places for water to escape from. 11. The number of stomata. 12. The number of stomata increases, so there are more places for water to escape from.	2 Maximum two marks for structural features only with no explanation.

QUESTION

38 · 5090/22/O/N/22 Q8

- 8 (a) Describe and explain the gas exchange that takes place between the leaf of a plant and the air in the atmosphere during a 24-hour period.
- (b) Outline the movement of water through a leaf during the process of transpiration.
- 9 (a) Explain the advantages and disadvantages of the use of insecticides in agriculture.

MARK SCHEME

5090/22/O/N/22 Q8

8(a)	<i>day</i> 1 light / day + photosynthesis ; 2 faster than respiration / respiration uses up oxygen ; 3 day + CO₂ used / in + O₂ produced / out ; <i>night</i> 4 no light / night + no photosynthesis ; 5 respiration ; 6 night / evening + O₂ used / in + CO₂ produced / out ; 7 stomata + open in day / closed at night ; 8 more water vapour lost during day ; 9 compensation point(s) AW ;	5	
8(b)	1 xylem ; 2 from xylem into mesophyll ; 3 diffusion / osmosis ; 4 down water potential gradient AW ; 5 layer of water + mesophyll + cell wall / outside cell ; 6 evaporation / water vapour ; 7 air / <u>inter</u>cellular + spaces ; 8 exits leaf through stomata / stoma ;	5	

QUESTION

39 · 5090/21/O/N/23 Q7

- 7 (a) A root hair cell in a plant is respiring aerobically.
- (i) Name the structures within the cell that carry out aerobic respiration.
.....
- (ii) Write the balanced symbol equation for the aerobic respiration of glucose, $C_6H_{12}O_6$.
.....
- (b) The plant experiences changing weather conditions including:
high temperature heavy rain high winds low humidity
- State the weather condition that may lead to a root hair cell respiring anaerobically and explain your answer.
weather condition
explanation
- 8 (a) The number of individuals in a population can vary with time.

MARK SCHEME

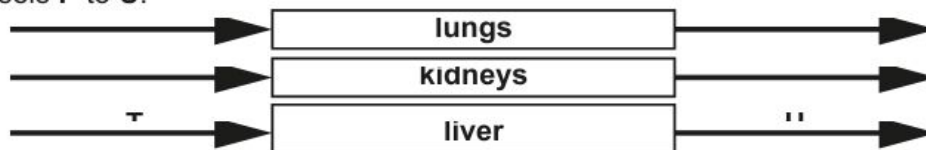
5090/21/O/N/23 Q7

7(a)(i)	mitochondria / mitochondrion ;	1	
7(a)(ii)	$C_6H_{12}O_6 + 6O_2$; $6H_2O + 6CO_2$;	2	A correct word equation for max 1 mark Ig any reference to energy or ATP
7(b)	heavy rain ; soil waterlogged / AW + no air spaces / oxygen available ;	2	

QUESTION

40 · 5090/22/O/N/23 Q3

- 3 Fig. 3.1 shows diagrams that represent the blood flow into and out of three major organs through blood vessels **P** to **U**.



- (a) Identify each of the following blood vessels.
- (b) Compare the structures of blood vessels **Q** and **R**.
- (c) Explain each of the following.
- (i) Blood in vessel **S** has a lower concentration of urea than blood in vessel **R**.
- (ii) Blood in vessel **U** sometimes has a higher concentration of glucose than blood in
- 4 Fig. 4.1 shows a Siamese fighting fish that lives in fresh water.

MARK SCHEME

5090/22/O/N/23 Q3

3(a)	(Q) <u>pulmonary vein</u> ; (R) <u>renal artery</u> ; (U) <u>hepatic vein</u> ;	3																
3(b)	<table><tr><td>Q – (pulmonary) vein</td><td></td><td>R – (renal) artery</td></tr><tr><td>thin + wall</td><td>+</td><td>thick + wall ;</td></tr><tr><td>thin muscle / thin elastic</td><td>+</td><td>thick muscle / thick elastic ;</td></tr><tr><td>wide + lumen AW</td><td>+</td><td>narrow + lumen AW ;</td></tr><tr><td>valves</td><td>+</td><td>no valves ;</td></tr></table>	Q – (pulmonary) vein		R – (renal) artery	thin + wall	+	thick + wall ;	thin muscle / thin elastic	+	thick muscle / thick elastic ;	wide + lumen AW	+	narrow + lumen AW ;	valves	+	no valves ;	3	/
Q – (pulmonary) vein		R – (renal) artery																
thin + wall	+	thick + wall ;																
thin muscle / thin elastic	+	thick muscle / thick elastic ;																
wide + lumen AW	+	narrow + lumen AW ;																
valves	+	no valves ;																
3(c)(i)	1 urea + into kidney OR kidney removes + urea ; 2 filtration / filtering ; 3 nephron / Bowman's capsule / glomerulus ; 4 urea + excreted / in urine ;	3																

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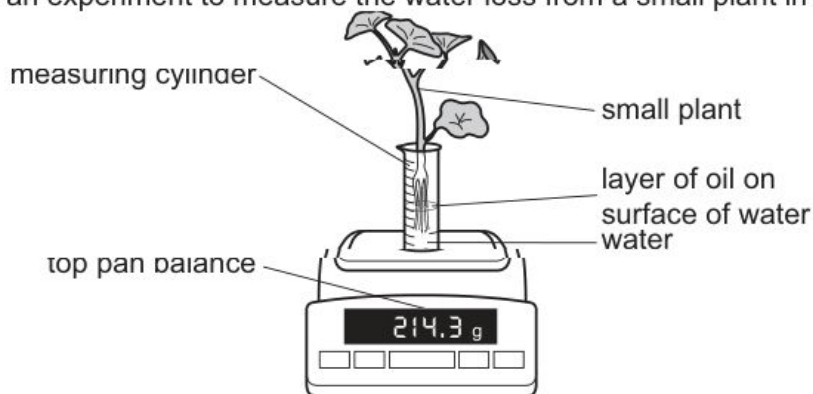
3(c)(ii)	1 (blood) glucose decreased / sugar low / body needs more sugar ; 2 glycogen ; 3 breakdown / hydrolysis + to glucose ; 4 glucose + passed out of liver / goes into U / hepatic vein / blood ; 5 (stimulated by) glucagon ;		
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Question	Answer	Marks	Guidance
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QUESTION

41 · 5090/21/O/N/24 Q1

- 1 Fig. 1.1 shows an experiment to measure the water loss from a small plant in 24 hours.



- (a) (i) Name the process by which a plant loses water from cells to the atmosphere.
 (ii) Suggest the purpose of the layer of oil.
 (iii) Describe **one** way in which a student could use this apparatus to obtain a value for the amount of water the plant loses in 24 hours.

- (b) Water is lost mainly through the plant's stomata.

Name **three** types of cell that the water passes through before it reaches the stomata in the boxes. Write the names of the types of cell in the correct order in the boxes.

_____ | → | _____ | → | _____ | → | _____ | → | stomata |

- (c) The experiment shown in Fig. 1.1 was repeated many times to investigate the effect of humidity on the rate of water loss.

All environmental conditions were kept constant except for humidity which was gradually changed. Fig. 1.2 shows the axes for a graph.

- (i) Sketch a graph on these axes of the expected results for this investigation. [1]
 (ii) Explain what your graph shows about the effect of humidity on water loss.

- 2 Maltase is an enzyme found in the intestines of vertebrates, including humans.

MARK SCHEME

5090/21/O/N/24 Q1

1(a)(i)	transpiration ;	1	
1(a)(ii)	prevent water loss / evaporation of water;	1	
1(a)(iii)	measure mass / volume at start + at end / after 24 hours ; subtract end value from start value / calculate difference ;	2	
1(b)	any three in correct sequence from: root hair (cell) ; (root) cortex (cell) ; xylem (vessel element) ; palisade / spongy / mesophyll (cell) ;	3	
1(c)(i)	line that is at its highest at a low humidity and at its lowest at a high humidity ;	1	
1(c)(ii)	the lower the humidity, the higher the rate of water loss AW ; water is lost by diffusion ; when humidity is low, the concentration of water outside will be lower than inside the leaf / atmosphere would not be saturated with water ;	3	

QUESTION

42 · 5090/21/O/N/24 Q4

- 4 Living organisms excrete carbon dioxide.
- (a) (i) Describe the meaning of the term excretion.
- (ii) Name the process which produces carbon dioxide in living organisms and state how carbon dioxide is excreted by humans.
- (b) The volume of carbon dioxide excreted varies over a 24-hour period and the factors that affect this are different for animals and plants. Suggest and explain reasons why the volume of carbon dioxide excreted varies for both animals and plants.
- (c) Explain the effect of an increasing concentration of carbon dioxide in the atmosphere.
- 5 The castor oil plant is a flowering plant which produces seeds.

MARK SCHEME

5090/21/O/N/24 Q4

4(a)(i)	any two from: removal + waste products / AW ; removal + toxic materials ; removal + substances made by the body / metabolism ;	2	
4(a)(ii)	<u>respiration</u> ; breathed out / exhaled / removed at the lungs / alveoli ;	2	

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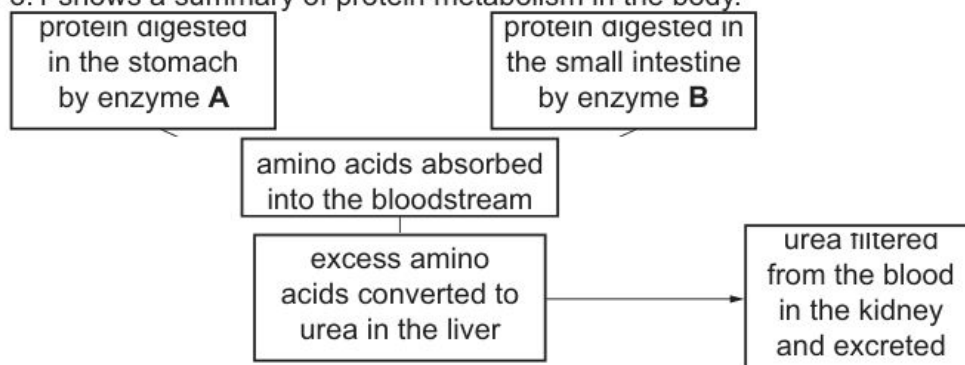
4(b)	<i>animals</i> activities vary / example given ; so more respiration at some times ; <i>plants</i> respiration + constant / ongoing ; daylight + photosynthesis AW ; carbon dioxide used for photosynthesis ;	4	max two marks for plants
4(c)	any two from: acts as a greenhouse gas ; traps thermal / heat energy / global warming ; climate change / specific effect on land or organisms ;	2	

Question	Answer	Marks	Guidance
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QUESTION

43 · 5090/21/O/N/25 Q3

- 3 (a) Fig. 3.1 shows a summary of protein metabolism in the body.



Use Fig. 3.1 to answer these questions.

- (i) State the name of enzyme **A** and enzyme **B**.
 (ii) State the name of blood vessel **C**.
 (iii) Name the capillary structures in the kidney that filter urea from the blood.
- (b) Two groups of students, **X** and **Y**, investigate the effect of different diets on the concentration of urea in their urine.
 For one day, the two groups eat different foods:
 • group **X** eat bread, jam, orange juice, lettuce, fruit
 • group **Y** eat nuts, milk, bread, cheese, meat.
 Early the next morning, they collect and analyse their urine to measure the concentration of urea.
 Table 3.1 shows their results.

	concentration of urea / g per 100 cm ³ of urine						
	individual students						mean
group X	1.5,	1.7,	2.3,	1.6,	2.1,	1.4	1.8
group Y	1.8,	1.9,	2.3,	2.5,	1.9,	2.4	

- (i) Calculate the mean concentration of urea for the students in group **Y**.
 Give your answer to **two** significant figures.
 Space for working.
 mean concentration of urea = g per 100 cm³ of urine [2]
- (ii) Explain the difference in the mean concentration of urea for the two groups of students.

- (iii) One of the students said: 'To make a fair comparison, we should have made sure that the students in each group drank the same volume of water'.
 Explain why this student is correct.

- 4 Chlorophyll is found in all organisms that photosynthesise.

MARK SCHEME

5090/21/O/N/25 Q3

3(a)(i)	pepsin ; trypsin ;	2	
3(a)(ii)	hepatic portal vein ;	1	
3(a)(iii)	glomerulus ;	1	
3(b)(i)	2.1 ;	2	A 12.8 / 6 OR 2.133 = 1 mark
3(b)(ii)	group Y ate more protein / ate protein rich foods ; so / in group Y, more amino acids are deaminated to produce urea ; so / in group Y, more urea is filtered / passes into the urine in the kidney ;	3	A ora for group X
3(b)(iii)	idea that the volume of water drunk, will affect the volume of water used to form urine ; this / the volume of water drunk, will affect the concentration of urea in the urine ;	2	

Question	Answer	Marks	Guidance
----------	--------	-------	----------

QUESTION

44 · 5090/21/O/N/25 Q4

- (b) Mistletoe is a parasitic green plant that can often be seen growing attached to trees. Fig. 4.1 shows some mistletoe growing on the branch of a tree.



Most plants have about 1 mg of chlorophyll in 100 g of their leaves. Mistletoe, however, has about 0.4 mg of chlorophyll in 100 g of its leaves. Mistletoe seeds are stuck to the branches of trees by birds. The seeds germinate to produce special roots that grow into the phloem of the tree branch. Suggest why mistletoe grows these special roots.

- 5 (a) Fig. 5.1 shows a model of the human circulatory system.

MARK SCHEME

5090/21/O/N/25 Q4

4(a)	magnesium ; chloroplasts ; palisade (mesophyll) ; sucrose ; cellulose ;	5	
------	---	---	--

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Page 9 of 13

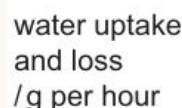
4(b)	max four from: mistletoe / it has less chlorophyll ; so can perform less photosynthesis ; therefore, it makes less sugars ; so the mistletoe obtains sugars / sucrose from the tree ; phloem contains sugars / sucrose ;	4	
------	--	---	--

Question	Answer	Marks	Guidance
----------	--------	-------	----------

QUESTION

45 · 5090/22/O/N/25 Q2

- At 0 hours it was sunrise and at 15 hours it was sunset. The results are shown in Fig. 2.2.



- experiment.

- [Total: 12]

-

MARK SCHEME

5090/22/O/N/25 02

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Question	Answer	Marks	Guidance
----------	--------	-------	----------

QUESTION

46 · 5090/11/M/J/20 Q2

2 By which process does water vapour pass out of a leaf?

- A active transport
B diffusion
C translocation

MARK SCHEME

5090/11/M/J/20 Q2

2

B

1

QUESTION

47 · 5090/11/M/J/20 Q11

11 Which feature of root hairs suggests that they take up ions from the soil by active transport?

- A Their cell membranes are partially permeable.
B They have a large surface area.
C They have a lower water potential than the soil.
D They take up ions more slowly in low oxygen concentrations.

12 The photomicrograph shows part of a section through a root.

MARK SCHEME

5090/11/M/J/20 Q11

11

D

1

QUESTION

48 · 5090/11/M/J/20 Q19

19 The diagram shows a kidney and blood vessels associated with it.



Which statements are correct?

- 1 vessel X contains more urea than vessel Y.
2 vessel Y contains urine.
3 vessel X contains more oxygen than vessel Y.

A 1, 2 and 3 B 1 and 3 only C 1 only D 2 and 3 only

20 Which of these statements describes control by **negative** feedback?

MARK SCHEME

5090/11/M/J/20 Q19

19

B

1

QUESTION

49 · 5090/12/M/J/20 Q2

2 Which movement of a substance in plants is an energy-consuming process?

- A absorption of carbon dioxide by a palisade cell
B absorption of oxygen by a mesophyll cell
C nitrate uptake by root hair cells
D transport of water up through the xylem

MARK SCHEME

5090/12/M/J/20 Q2

2

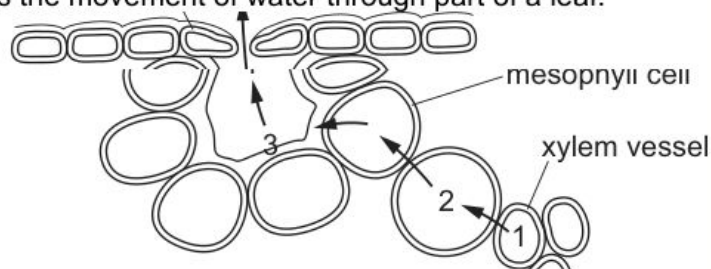
C

1

QUESTION

50 · 5090/12/M/J/20 Q3

- 3 The diagram shows the movement of water through part of a leaf.



Which processes are involved in the movement of water at these stages?

	1 → 2	2 → 3	3 → 4
A	diffusion	evaporation	osmosis
B	evaporation	diffusion	osmosis
C	osmosis	diffusion	evaporation
D	osmosis	evaporation	diffusion

- 4 Which substance is an enzyme?

MARK SCHEME

5090/12/M/J/20 Q3

3	D	1
---	---	---

QUESTION

51 · 5090/12/M/J/20 Q5

- 5 What is the effect on plants of a lack of magnesium ions?

- A Chlorophyll production is reduced.
 B Leaves turn dark green.
 C Starch production increases.
 D Young leaves wilt.

- 6 The diagram shows a green plant photosynthesising under a glass jar. This was used as a

MARK SCHEME

5090/12/M/J/20 Q5

5	A	1
---	---	---

QUESTION

52 · 5090/12/M/J/20 Q11

- 11 Water and ions can reach the xylem of a plant root through cell walls, without passing through a cell membrane.

How do these substances move through the cell walls?

	water	ions
A	diffusion	diffusion
B	diffusion	osmosis
C	osmosis	diffusion
D	osmosis	osmosis

- 12 The photomicrograph shows part of a section through a root.

MARK SCHEME

5090/12/M/J/20 Q11

11	A	1
----	---	---

QUESTION

53 · 5090/12/M/J/20 Q19

19 The diagram shows the human kidneys and their associated structures.



What is the structure labelled **x**?

- A** renal artery
B renal vein

MARK SCHEME

5090/12/M/J/20 Q19

19

C

1

QUESTION

54 · 5090/11/O/N/20 Q6

6 How do carbon dioxide and water enter a leaf?

	carbon dioxide	water
A	diffusion	active transport
B	diffusion	transpiration pull
C	osmosis	active transport
D	osmosis	transpiration pull

MARK SCHEME

5090/11/O/N/20 Q6

6

B

1

QUESTION

55 · 5090/11/O/N/20 Q9

9 Blood flows away from the small intestine in the hepatic portal vein. Where does the blood go next, and what may happen there?

- A** the kidneys, delivering glucose for respiration
B the kidneys, for removal of urea
C the liver, where alcohol is broken down
D the liver, where bile is added to the blood

MARK SCHEME

5090/11/O/N/20 Q9

9

C

1

QUESTION

56 · 5090/11/O/N/20 Q11

11 Which diagram shows the position of phloem tissue in a herbaceous, dicotyledonous root?



MARK SCHEME

5090/11/O/N/20 Q11

11

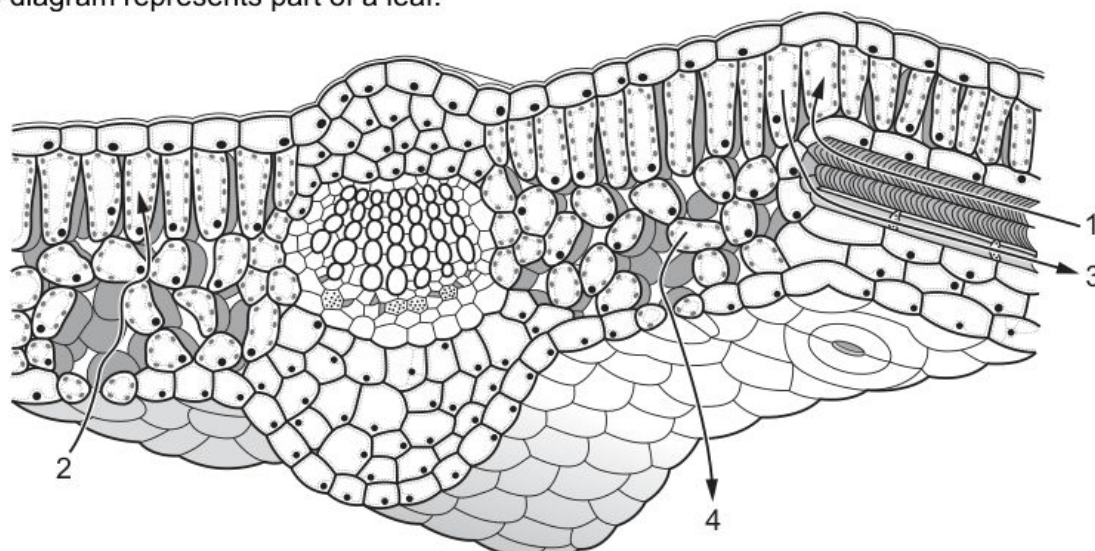
D

1

QUESTION

57 · 5090/11/O/N/20 Q12

12 The diagram represents part of a leaf.



Which arrows show translocation and transpiration?

	translocation	transpiration
☐	☐	☐

13 In the process of blood clotting, the1..... help to convert2..... to3..... which

MARK SCHEME

5090/11/O/N/20 Q12

12	C	1
----	---	---

QUESTION

58 · 5090/12/O/N/20 Q5

5 Which part of a leaf absorbs and uses carbon dioxide from the air?

B mesophyll

MARK SCHEME

5090/12/O/N/20 Q5

5	B	1
---	---	---

QUESTION

59 · 5090/12/O/N/20 Q6

6 How do carbon dioxide and water enter a leaf?

	carbon dioxide	water
A	diffusion	active transport
B	diffusion	transpiration pull
C	osmosis	active transport
D	osmosis	transpiration pull

MARK SCHEME

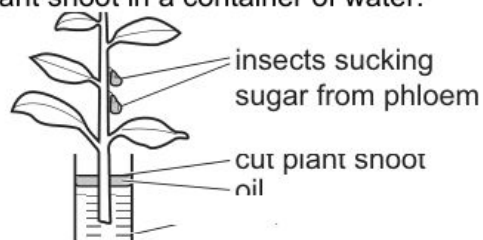
5090/12/O/N/20 Q6

6	B	1
---	---	---

QUESTION

60 · 5090/12/O/N/20 Q12

12 The diagram shows a cut plant shoot in a container of water.



What will stop the movement of water up this stem?

- A** a fall in the humidity of the air
B a rise in the air temperature
C a blockage in the xylem
D an insect sucking sugar from the phloem

MARK SCHEME

5090/12/O/N/20 Q12

12	C	1
----	---	---

QUESTION

61 · 5090/11/M/J/21 Q11

11 Which feature of a root hair cell helps it take up ions from water in the soil?

- A** The cell has a partially permeable cell wall.
B The cell is able to provide a source of energy.
C The cell vacuole has a high ion concentration.
D The cell vacuole has a low water potential.

MARK SCHEME

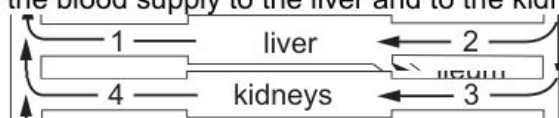
5090/11/M/J/21 Q11

11	B	1
----	---	---

QUESTION

62 · 5090/11/M/J/21 Q15

15 The diagram represents the blood supply to the liver and to the kidneys.



Which vessels contain blood with the highest and lowest concentrations of urea?

	highest	lowest
	1	4

MARK SCHEME

5090/11/M/J/21 Q15

15	B	1
----	---	---

QUESTION

63 · 5090/12/M/J/21 Q1

1 Which row is correct?

	cell	tissue	organ	organ system
A	chloroplast	mesophyll	liver	digestive
B	sap vacuole	red blood cell	blood	nervous
C	red blood cell	sap vacuole	stomach	liver
D	neurone	vascular bundle	leaf	excretory

MARK SCHEME

5090/12/M/J/21 Q1

1	D	1
---	---	---

QUESTION

64 · 5090/12/M/J/21 Q7

7 A lack of certain mineral ions affects the growth of young plants.
Which effects are caused by a lack of magnesium ions and a lack of nitrate ions?

	lack of magnesium ions	lack of nitrate ions
A	the leaves become dark green	overall growth is stunted
B	the leaves go yellow in patches	the plant is unable to form proteins
C	the plant is unable to form enzymes	the plant is unable to form starch
D	the plant is unable to form root hairs	the leaves become bright green

MARK SCHEME

5090/12/M/J/21 Q7

7	B	1
---	---	---

QUESTION

65 · 5090/12/M/J/21 Q11

11 How does water enter root hair cells?
A from high to low water potential by active transport
B from high to low water potential by osmosis
C from low to high water potential by active transport
D from low to high water potential by osmosis

MARK SCHEME

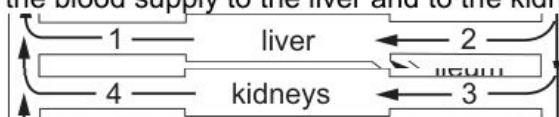
5090/12/M/J/21 Q11

11	B	1
----	---	---

QUESTION

66 · 5090/12/M/J/21 Q15

15 The diagram represents the blood supply to the liver and to the kidneys.



Which vessels contain blood with the highest and lowest concentrations of urea?

	highest	lowest
A	1	4

16 Which statement is true for aerobic respiration?

MARK SCHEME

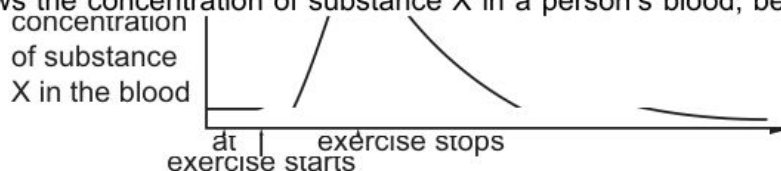
5090/12/M/J/21 Q15

15	B	1
----	---	---

QUESTION

67 · 5090/12/M/J/21 Q17

17 The graph shows the concentration of substance X in a person's blood, before, during and after



What is substance X?

B glycogen

MARK SCHEME

5090/12/M/J/21 Q17

17	C	1
----	---	---

QUESTION

68 · 5090/11/O/N/21 Q2

2 Which process occurs by diffusion alone?

- A** movement of carbon dioxide into the palisade cells in a leaf
- B** uptake of magnesium ions by root hair cells
- C** movement of food from the stomach to the duodenum
- D** transfer of glucose molecules from the ileum into the blood

3 An experiment to investigate osmosis is set up as shown.

MARK SCHEME

5090/11/O/N/21 Q2

2	A	1
---	---	---

QUESTION

69 · 5090/11/O/N/21 Q11

11 The diagram shows a section through a plant stem.



Which process is responsible for moving substances in region Z?

- A** diffusion
- B** osmosis
- C** translocation
- D** transpiration

12 Which statement describes why a leaf wilts?

MARK SCHEME

5090/11/O/N/21 Q11

11	C	1
----	---	---

QUESTION

70 · 5090/11/O/N/21 Q12

12 Which statement describes why a leaf wilts?

- A** Less water is absorbed by the root hairs when the stomata are closed.
- B** Less water is absorbed by the root hairs than is lost through the stomata.
- C** More water is absorbed by the root hairs when the stomata are closed.
- D** More water is absorbed by the root hairs than is lost through the stomata.

MARK SCHEME

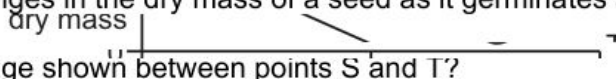
5090/11/O/N/21 Q12

12	B	1
----	---	---

QUESTION

71 · 5090/11/O/N/21 Q34

34 The graph shows changes in the dry mass of a seed as it germinates and grows.



What causes the change shown between points S and T?

- A** photosynthesis
- B** respiration
- C** transpiration

35 The graph shows the concentration in the blood of three of the four hormones FSH, LH,

MARK SCHEME

5090/11/O/N/21 Q34

34

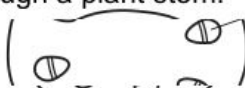
B

1

QUESTION

72 · 5090/12/O/N/21 Q11

11 The diagram shows a section through a plant stem.



Which process is responsible for moving substances in region Z?

- A** diffusion
- B** osmosis
- C** translocation
- D** transpiration

MARK SCHEME

5090/12/O/N/21 Q11

11

C

1

QUESTION

73 · 5090/12/O/N/21 Q12

12 What is transpiration?

- A** evaporation of water from the stomata
- B** evaporation of water at the surface of mesophyll cells and its loss through the stomata
- C** the movement of water through the cuticle
- D** the movement of water in the xylem from the roots to the leaves

MARK SCHEME

5090/12/O/N/21 Q12

12

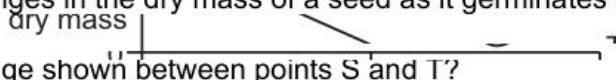
B

1

QUESTION

74 · 5090/12/O/N/21 Q34

34 The graph shows changes in the dry mass of a seed as it germinates and grows.



What causes the change shown between points S and T?

- A** photosynthesis
- B** respiration
- C** transpiration

MARK SCHEME

5090/12/O/N/21 Q34

34

B

1

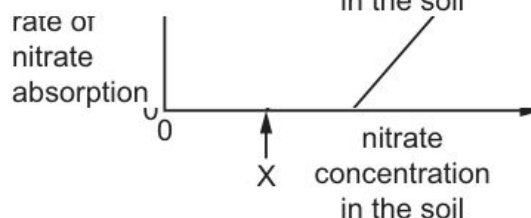
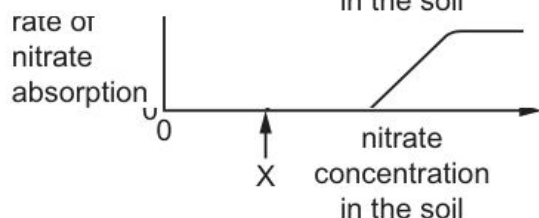
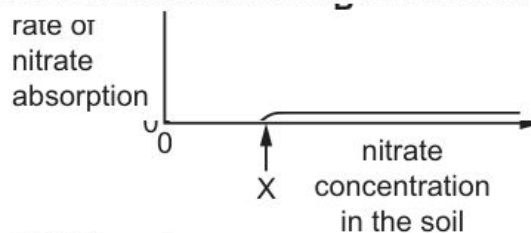
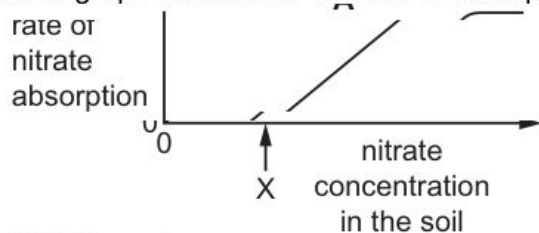
QUESTION

75 · 5090/11/M/J/22 Q4

- 4 The rate of nitrate ion absorption by a root hair cell was measured at different soil nitrate concentrations.

At X, the concentration of nitrate in the soil is the same as in the cell.

Which graph shows how the rate of absorption varies with nitrate concentration in the soil?



- 5 The graph shows how the rate of reaction of a protease from the stomach varies with pH.

MARK SCHEME

5090/11/M/J/22 Q4

4	A	1
---	---	---

QUESTION

76 · 5090/11/M/J/22 Q13

- 13 What is the main cause of water moving up to the leaves in xylem vessels?

- A active transport
B evaporation from the epidermis or the leaf
C evaporation from the walls of the mesophyll cells
D use of water in photosynthesis

- 14 Which blood vessels carry blood towards the heart and towards the lungs?

MARK SCHEME

5090/11/M/J/22 Q13

13	C	1
----	---	---

QUESTION

77 · 5090/12/M/J/22 Q2

- 2 Xylem vessels are cells that have become adapted for conduction and support. Which two adaptations assist them in these functions?

- A presence of a nucleus and cytoplasm
B lack of cytoplasm and woody cell walls
C lack of a nucleus and presence of cytoplasm
D presence of cytoplasm and woody cell walls

MARK SCHEME

5090/12/M/J/22 Q2

2	B	1
---	---	---

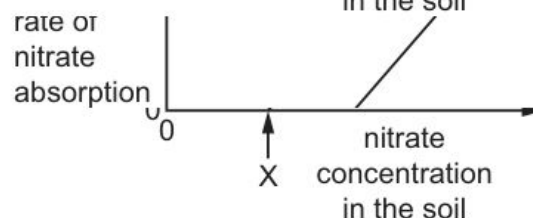
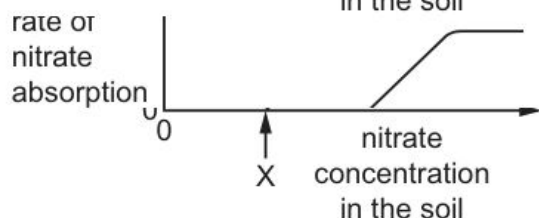
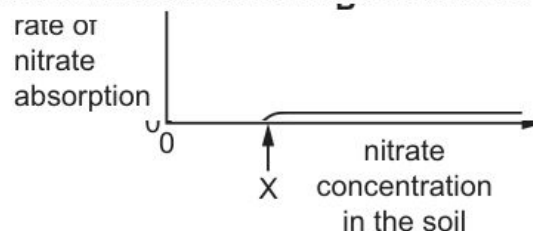
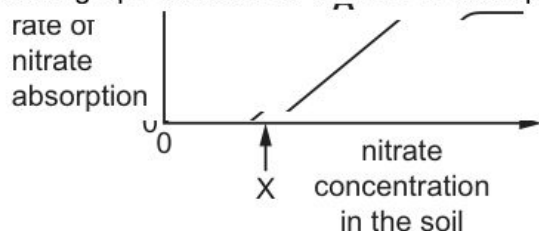
QUESTION

78 · 5090/12/M/J/22 Q4

- 4 The rate of nitrate ion absorption by a root hair cell was measured at different soil nitrate concentrations.

At X, the concentration of nitrate in the soil is the same as in the cell.

Which graph shows how the rate of absorption varies with nitrate concentration in the soil?



- 5 Amylase is an enzyme that breaks down starch to maltose.

MARK SCHEME

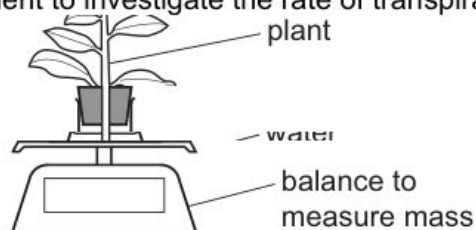
5090/12/M/J/22 Q4

4	A	1
---	---	---

QUESTION

79 · 5090/12/M/J/22 Q12

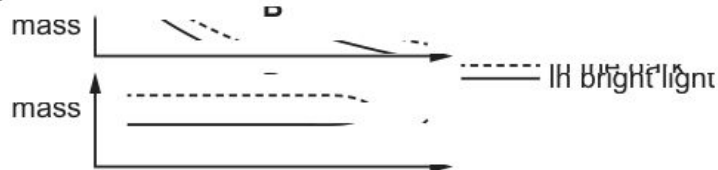
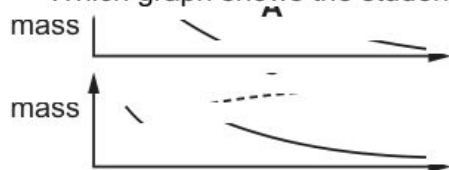
- 12 A student set up an experiment to investigate the rate of transpiration in a plant.



The student set up two identical sets of apparatus.

One was placed in a room at 20 °C with bright light and one in a room at 20 °C in the dark.

Which graph shows the student's results?



MARK SCHEME

5090/12/M/J/22 Q12

12	A	1
----	---	---

QUESTION

80 · 5090/12/M/J/22 Q13

- 13 What is the main cause of water moving up to the leaves in xylem vessels?
A active transport
B evaporation from the epidermis of the leaf
C evaporation from the walls of the mesophyll cells
D use of water in photosynthesis
- 14 Which blood vessels, linked directly to the heart, transport deoxygenated blood?

MARK SCHEME

5090/12/M/J/22 Q13

13	C	1
----	---	---

QUESTION

81 · 5090/11/O/N/22 Q2

- 2 What are the functions of xylem vessels?

	strengthening roots and stems	supporting leaf shape	transporting sugars
☐	☒	☒	☒

xylem vessels

MARK SCHEME

5090/11/O/N/22 Q2

2	B	1
---	---	---

QUESTION

82 · 5090/11/O/N/22 Q4

- 4 Which row applies to active transport?

	movement occurs from region of higher concentration to region of lower concentration	movement occurs from region of lower concentration to region of higher concentration	an example is ion uptake by root hairs	an example is water uptake by root hairs
☐	☒	☐	☐	☐

MARK SCHEME

5090/11/O/N/22 Q4

4	C	1
---	---	---

QUESTION

83 · 5090/11/O/N/22 Q11

- 11 Which function is **not** carried out by the liver?
A breakdown of alcohol
B conversion of glycogen to glucose
C formation of urea
D secretion of digestive enzymes

MARK SCHEME

5090/11/O/N/22 Q11

11	D	1
----	---	---

QUESTION

84 · 5090/11/O/N/22 Q12

12 Which statements about root hairs are correct?

- 1 The cell membrane can allow movement of water and ions into the cell.
- 2 The cell vacuole extends into the root hair.
- 3 They are formed as an extension of the outer layer of root cells.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

13 The graph shows the transpiration rates of two plants during one day. Both plants were grown

MARK SCHEME

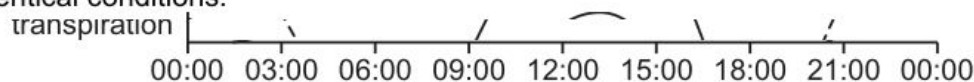
5090/11/O/N/22 Q12

12	A	1
----	---	---

QUESTION

85 · 5090/11/O/N/22 Q13

13 The graph shows the transpiration rates of two plants during one day. Both plants were grown under identical conditions.



Which statement explains what the graph shows?

- A** Plant X closes its stomata during the brightest part of the day.
B Plant Y closes its stomata during the brightest part of the day.
C Plants X and Y close their stomata during the brightest part of the day.
D Plant Y has no stomata.

MARK SCHEME

5090/11/O/N/22 Q13

13	B	1
----	---	---

QUESTION

86 · 5090/11/O/N/22 Q20

20 How do the concentrations of oxygen and urea in the renal artery compare to their concentrations in the renal vein?

	oxygen in renal artery	urea in renal artery
A	higher	higher
B	higher	lower
C	lower	higher
D	lower	lower

21 Which changes occur when a person walks from a very cold room into a hot room?

MARK SCHEME

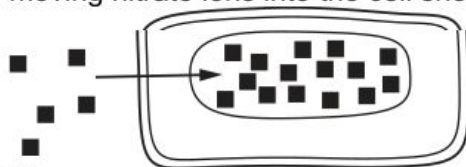
5090/11/O/N/22 Q20

20	A	1
----	---	---

QUESTION

87 · 5090/12/O/N/22 Q4

4 Which process is moving nitrate ions into the cell shown?



key
■ nitrate ion

- A active transport
B diffusion
C translocation

MARK SCHEME

5090/12/O/N/22 Q4

4

A

1

QUESTION

88 · 5090/12/O/N/22 Q11

11 Which function is **not** carried out by the liver?

- A breakdown of alcohol
B conversion of glycogen to glucose
C formation of urea
D secretion of digestive enzymes

12 Which statements about root hairs are correct?

- 1 The cell membrane can allow movement of water and ions into the cell.
2 The cell vacuole extends into the root hair.
3 They are formed as an extension of the outer layer of root cells.

- A 1, 2 and 3 B 1 and 2 only C 1 and 3 only D 2 and 3 only

MARK SCHEME

5090/12/O/N/22 Q11

11

D

1

QUESTION

89 · 5090/12/O/N/22 Q12

12 Which statements about root hairs are correct?

- 1 The cell membrane can allow movement of water and ions into the cell.
2 The cell vacuole extends into the root hair.
3 They are formed as an extension of the outer layer of root cells.

- A 1, 2 and 3 B 1 and 2 only C 1 and 3 only D 2 and 3 only

MARK SCHEME

5090/12/O/N/22 Q12

12

A

1

QUESTION

90 · 5090/12/O/N/22 Q13

13 The table shows weather conditions in a field.
Which row shows conditions that would cause a plant to wilt most rapidly?

	air temperature /°C	humidity	time of day	wind speed /km per hour
A	15	raining	00:00	32
B	12	dry	14:00	24
C	6	fog/rain	08:00	3
D	5	dry	22:00	16

MARK SCHEME

5090/12/O/N/22 Q13

13

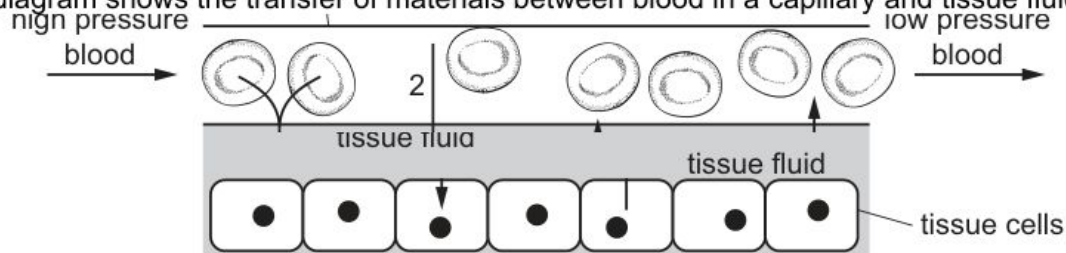
B

1

QUESTION

91 · 5090/12/O/N/22 Q16

16 The diagram shows the transfer of materials between blood in a capillary and tissue fluid.



Which row shows the correct labels for the arrows in the diagram?

A	carbon dioxide	glucose	oxygen	water
B	glucose	water	oxygen	urea
C	oxygen	glucose	carbon dioxide	water
D	oxygen	water	carbon dioxide	glucose

17 An experiment is set up, as shown.

MARK SCHEME

5090/12/O/N/22 Q16

16

C

1

QUESTION

92 · 5090/12/O/N/22 Q20

20 What are two main functions of the kidneys?

- A to make toxins in the blood harmless and to remove excess salt in the blood
- B to remove excess salt in the blood and to remove urea from the blood plasma
- C to remove urea from the blood plasma and to convert glucose to glycogen
- D to convert glucose to glycogen and to make toxins in the blood harmless

MARK SCHEME

5090/12/O/N/22 Q20

20

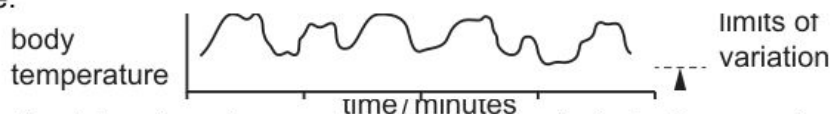
B

1

QUESTION

93 · 5090/12/O/N/22 Q22

22 The graph shows measurements taken of one aspect of the internal environment of the human body over time.



What controls the internal environment of the human body to keep each aspect within strict

- A excretion
- B homeostasis
- C positive feedback
- D respiration

MARK SCHEME

5090/12/O/N/22 Q22

22

B

1

QUESTION

94 · 5090/11/M/J/23 Q9

- 9 Which description of the function of root hairs is correct?
- A** Root hairs absorb ions by active transport and osmosis.
B Root hairs absorb ions by osmosis only.
C Root hairs absorb water by active transport.
D Root hairs absorb water by osmosis.

MARK SCHEME

5090/11/M/J/23 Q9

9	D	1
---	---	---

QUESTION

95 · 5090/11/M/J/23 Q10

- 10 Which conditions result in the highest rate of transpiration from a plant?

	percentage humidity	temperature /°C
A	88	22

- 11 Which disease can be the result of a dietary deficiency of both a mineral ion and a vitamin?

MARK SCHEME

5090/11/M/J/23 Q10

10	B	1
----	---	---

QUESTION

96 · 5090/11/M/J/23 Q12

- 12 A student states: 'The contraction of circular muscles pushes the contents forward.' Which process is being described?
- A** peristalsis
B deamination
C translocation
D transpiration

MARK SCHEME

5090/11/M/J/23 Q12

12	A	1
----	---	---

QUESTION

97 · 5090/11/M/J/23 Q23

- 23 The blood in the renal vein has a different composition to the blood in the renal artery. How is the blood in the renal vein different?
- A** less carbon dioxide, less urea, less water
B less oxygen, more urea, less water
C more carbon dioxide, less urea, less water
D more oxygen, less urea, less water

MARK SCHEME

5090/11/M/J/23 Q23

23	C	1
----	---	---

QUESTION

98 · 5090/12/M/J/23 Q9

- 9 Root hairs take in water and ions from the soil. Which table shows the correct concentrations of water and ions in the root hair?

substance	concentration in root hair
water	lower than the soil
ions	higher than the soil

substance	concentration in root hair
water	higher than the soil
ions	lower than the soil

substance	concentration in root hair
water	higher than the soil
ions	higher than the soil

substance	concentration in root hair
water	lower than the soil
ions	lower than the soil

- 10 A herbaceous plant, growing in a nutrient solution, is placed in a well-lit container. Humid air is

MARK SCHEME

5090/12/M/J/23 Q9

9	A	1
---	---	---

QUESTION

99 · 5090/12/M/J/23 Q23

- 23 In which structure is there a fluid with a low concentration of urea?



MARK SCHEME

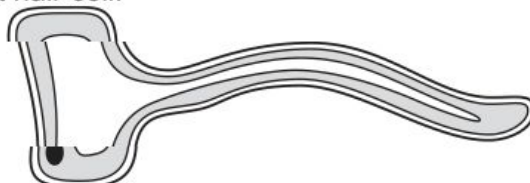
5090/12/M/J/23 Q23

23	A	1
----	---	---

QUESTION

100 · 5090/11/O/N/23 Q9

- 9 The diagram shows a root hair cell.



How does the shape of the cell help the plant?

- A It allows roots to search for ions.
 B It allows roots to search for water.
 C It increases the surface area for sugar uptake.
 D It increases the surface area for water uptake.
- 10 A student used the apparatus shown to measure the rate of water loss by the leafy stem at

MARK SCHEME

5090/11/O/N/23 Q9

9	D	1
---	---	---

QUESTION

101 · 5090/12/O/N/23 Q1

1 Which structures have both cytoplasm and cell walls?

- B** red blood cells
C root hair cells
D xylem vessels

MARK SCHEME

5090/12/O/N/23 Q1

1

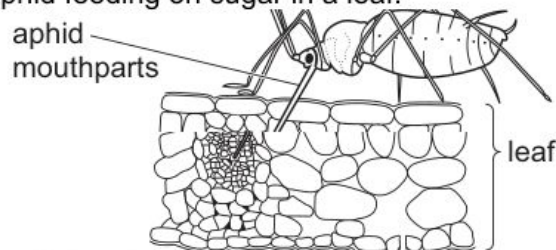
C

1

QUESTION

102 · 5090/12/O/N/23 Q7

7 The diagram shows an aphid feeding on sugar in a leaf.



When the aphid feeds, in which order do its mouthparts pass through the tissues of the leaf?

- A** palisade cells → phloem → upper epidermis
B upper epidermis → palisade cells → phloem
C palisade cells → upper epidermis → xylem
D upper epidermis → xylem → palisade cells

MARK SCHEME

5090/12/O/N/23 Q7

7

B

1

QUESTION

103 · 5090/12/O/N/23 Q9

9 Which row shows the processes involved when nitrate ions and water, from soil, enter a root hair

	nitrate ions	water
A	active transport	osmosis
B	diffusion	osmosis
C	diffusion	active transport
D	osmosis	diffusion

10 Which change in conditions will decrease the rate of transpiration?

MARK SCHEME

5090/12/O/N/23 Q9

9

A

1

QUESTION

104 · 5090/12/O/N/23 Q10

10 Which change in conditions will decrease the rate of transpiration?

- A** a decrease in air currents around the leaf
B a decrease in the humidity of the atmosphere
C an increase in light intensity around the leaf
D an increase in the temperature of the atmosphere

MARK SCHEME

5090/12/O/N/23 Q10

10

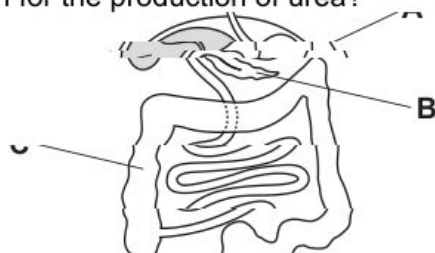
A

1

QUESTION

105 · 5090/12/O/N/23 Q23

23 Which label shows the location for the production of urea?



24 A student was asked to compare a typical spinal reflex action with a voluntary action.

MARK SCHEME

5090/12/O/N/23 Q23

23

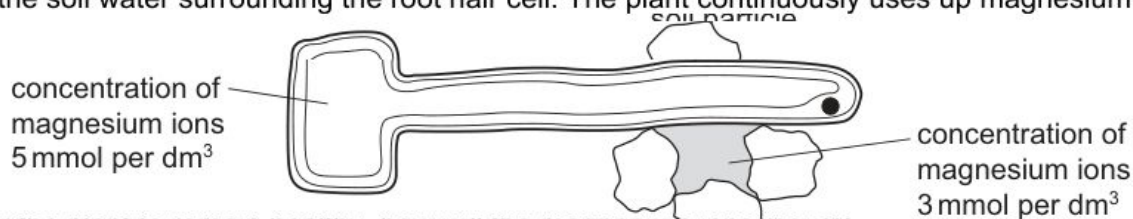
D

1

QUESTION

106 · 5090/11/M/J/24 Q4

4 The diagram shows the concentration of magnesium ions in a root hair cell of a healthy plant and in the soil water surrounding the root hair cell. The plant continuously uses up magnesium ions.



For the plant to remain healthy, how will the magnesium ions move?

- A into the cell by active transport
- B into the cell by diffusion
- C out of the cell by active transport
- D out of the cell by diffusion

MARK SCHEME

5090/11/M/J/24 Q4

4

A

1

QUESTION

107 · 5090/11/M/J/24 Q9

9 Which pathway is taken by water and mineral ions through a plant?

- A root hair cells → mesophyll cells → xylem → cortex cells
- B root hair cells → cortex cells → phloem → mesophyll cells
- C root hair cells → mesophyll cells → phloem → cortex cells
- D root hair cells → cortex cells → xylem → mesophyll cells

10 A student investigated the rate of transpiration in a green plant. The plant was growing in soil in

MARK SCHEME

5090/11/M/J/24 Q9

9

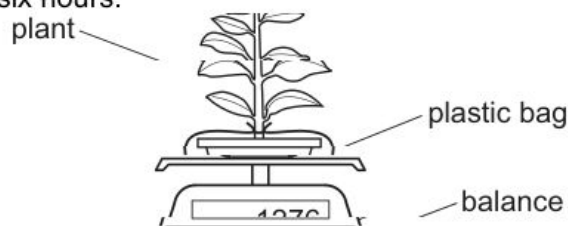
D

1

QUESTION

108 · 5090/11/M/J/24 Q10

- 10 A student investigated the rate of transpiration in a green plant. The plant was growing in soil in the pot. The student watered the plant and then put a plastic bag around the pot, leaving the leaves of the plant outside the bag. She measured the mass of the plant in the pot at the start and repeated this every two hours for six hours.



The results of the investigation are shown.

time of day	mass of pot and plant / g
-------------	---------------------------

What is the mean rate of transpiration over the six-hour period?

- A 43 g/h B 12 g/h C 8 g/h D 4 g/h
- 11 The diagram shows what happens to amino acids in part of the human body.

MARK SCHEME

5090/11/M/J/24 Q10

10	D	1
----	---	---

QUESTION

109 · 5090/11/M/J/24 Q13

- 13 A healthy person eats a meal containing carbohydrates. Two hours later, which structure will contain a liquid with an increased concentration of glucose?
- B hepatic portal vein
C renal artery

MARK SCHEME

5090/11/M/J/24 Q13

13	B	1
----	---	---

QUESTION

110 · 5090/11/M/J/24 Q22

- 22 Which row identifies the contents of urine in a healthy person?

	glucose	urea	water	ions
A	✓	✓	✓	✓

MARK SCHEME

5090/11/M/J/24 Q22

22	D	1
----	---	---

QUESTION

111 · 5090/11/M/J/24 Q26

- 26 Which process is stimulated by adrenaline in the cells of the liver?
- A breakdown of glycogen, increasing the blood glucose level
B breakdown of excess amino acids, forming urea
C breakdown of proteins, releasing amino acids into the blood
D conversion of excess blood glucose to glycogen

MARK SCHEME

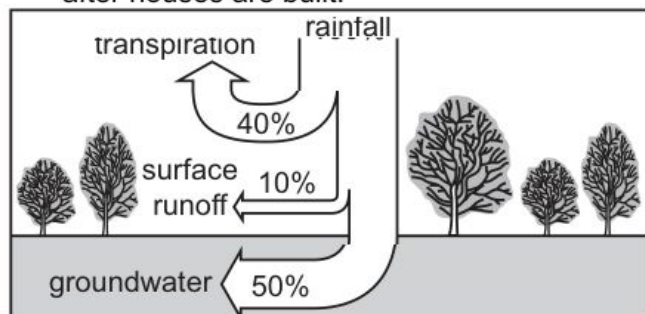
5090/11/M/J/24 Q26

26	A	1
----	---	---

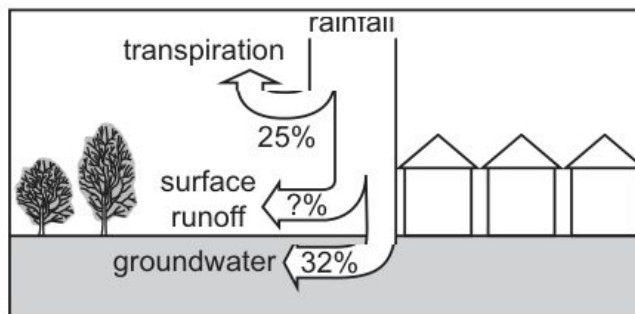
QUESTION

112 · 5090/11/M/J/24 Q39

- 39** The diagrams show what happens to rain when it falls on an area of forested land before and after houses are built.



before houses are built



after houses are built

What is the difference in the percentage of surface runoff before and after building the houses?

- A** increases by 33%
B increases by 43%
C decreases to 43%
D decreases to 33%

MARK SCHEME

5090/11/M/J/24 Q39

39

A

1

QUESTION

113 · 5090/12/M/J/24 Q9

- 9** Water moves from the soil to the atmosphere through a plant. The water enters the plant through the root hair cells and moves into the root cortex. Where does the water move into from the root cortex?

- A** into the mesophyll cells
B into the phloem cells
C into the xylem vessels
D through the stomata

- 10** The graph shows the loss of water vapour from two different plants growing in identical

MARK SCHEME

5090/12/M/J/24 Q9

9

C

1

QUESTION

114 · 5090/12/M/J/24 Q21

- 21** What is an example of excretion?
A release of a hormone into the blood
B removal of carbon dioxide from the lungs
C removal of undigested food from the digestive system
D release of water from the sweat glands

MARK SCHEME

5090/12/M/J/24 Q21

21

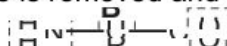
B

1

QUESTION

115 · 5090/12/M/J/24 Q22

- 22 The diagram shows an amino acid molecule.
Which part of the amino acid molecule is removed and used to make urea?



- 23 What is a function of sensory neurones?

MARK SCHEME

5090/12/M/J/24 Q22

22	A	1
----	---	---

QUESTION

116 · 5090/12/M/J/24 Q38

- 38 Which process is involved in the conversion of ammonia to nitrates?

- A denitrification
B excretion
C nitrification
D nitrogen fixation

- 39 Cutting down tropical rainforest trees has many consequences.

MARK SCHEME

5090/12/M/J/24 Q38

38	C	1
----	---	---

QUESTION

117 · 5090/11/O/N/24 Q4

- 4 Which movement of a substance in a plant requires the plant to provide energy?

- A absorption of carbon dioxide by a palisade cell
B absorption of oxygen by a mesophyll cell
C nitrate uptake by root hair cells
D transport of water up through the xylem

MARK SCHEME

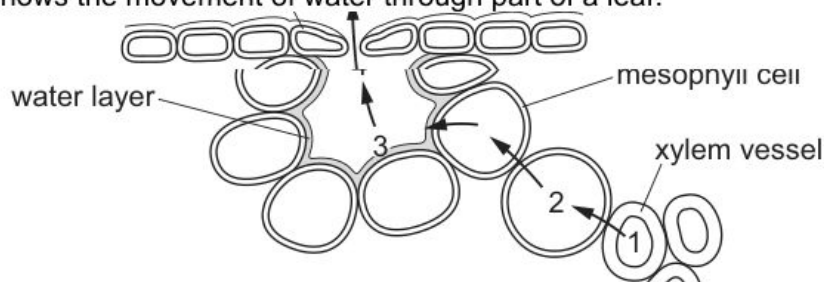
5090/11/O/N/24 Q4

4	C	1
---	---	---

QUESTION

118 · 5090/11/O/N/24 Q10

10 The diagram shows the movement of water through part of a leaf.



Which processes are involved in the movement of water at these stages?

	1 → 2	2 → 3	3 → 1
A	diffusion	evaporation	osmosis
B	evaporation	diffusion	osmosis
C	osmosis	diffusion	evaporation
D	osmosis	evaporation	diffusion

11 The diagram shows part of the human digestive system.

MARK SCHEME

5090/11/O/N/24 Q10

10	D	1
----	---	---

QUESTION

119 · 5090/11/O/N/24 Q15

15 Scientists have concluded that absorption of mineral ions in plants requires energy from respiration. Which observation best supports this conclusion?

- A Carbohydrate is stored in the roots.
- B Living roots give off carbon dioxide.
- C The root hairs have a large surface area.
- D Uptake of mineral ions is reduced in lower oxygen concentrations.

MARK SCHEME

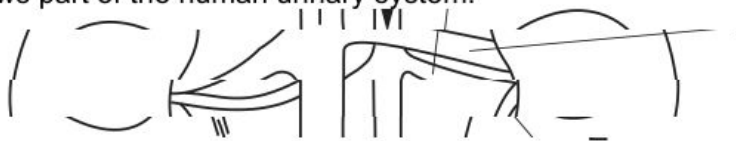
5090/11/O/N/24 Q15

15	D	1
----	---	---

QUESTION

120 · 5090/11/O/N/24 Q22

22 The diagram shows part of the human urinary system.



What are structures X, Y and Z?

	X	Y	Z
A	artery	vein	ureter
B	artery	vein	urethra
C	vein	artery	ureter
D	vein	artery	urethra

MARK SCHEME

5090/11/O/N/24 Q22

22	A	1
----	---	---

QUESTION

121 · 5090/12/O/N/24 Q1

1 The diagram shows three different types of microscopic structure.



What are these structures?

A	xylem vessel	root hair cell	red blood cell
B	red blood cell	xylem vessel	root hair cell
C	root hair cell	red blood cell	xylem vessel
D	xylem vessel	red blood cell	root hair cell

MARK SCHEME

5090/12/O/N/24 Q1

1	D	1
---	---	---

QUESTION

122 · 5090/12/O/N/24 Q9

9 Root hair cells take in water and ions from the soil.
Which row shows how water and ions are taken into root hair cells?

	water	ions
A	active transport	active transport
B	active transport	osmosis
C	osmosis	osmosis
D	osmosis	active transport

MARK SCHEME

5090/12/O/N/24 Q9

9	D	1
---	---	---

QUESTION

123 · 5090/12/O/N/24 Q22

22 The diagram shows the excretory system.



Which row identifies the names of the labelled structures?

A	bladder	ureter	kidney	urethra
B	bladder	urethra	kidney	ureter
C	kidney	ureter	bladder	urethra
D	kidney	urethra	bladder	ureter

MARK SCHEME

5090/12/O/N/24 Q22

22	C	1
----	---	---

QUESTION

124 · 5090/11/M/J/25 Q3

3 The diagram shows the structures found in cell X.

cytoplasm

cell membrane

Which type of cell is X?

- B** red blood cell
C root hair cell
D xylem vessel

MARK SCHEME

5090/11/M/J/25 Q3

3

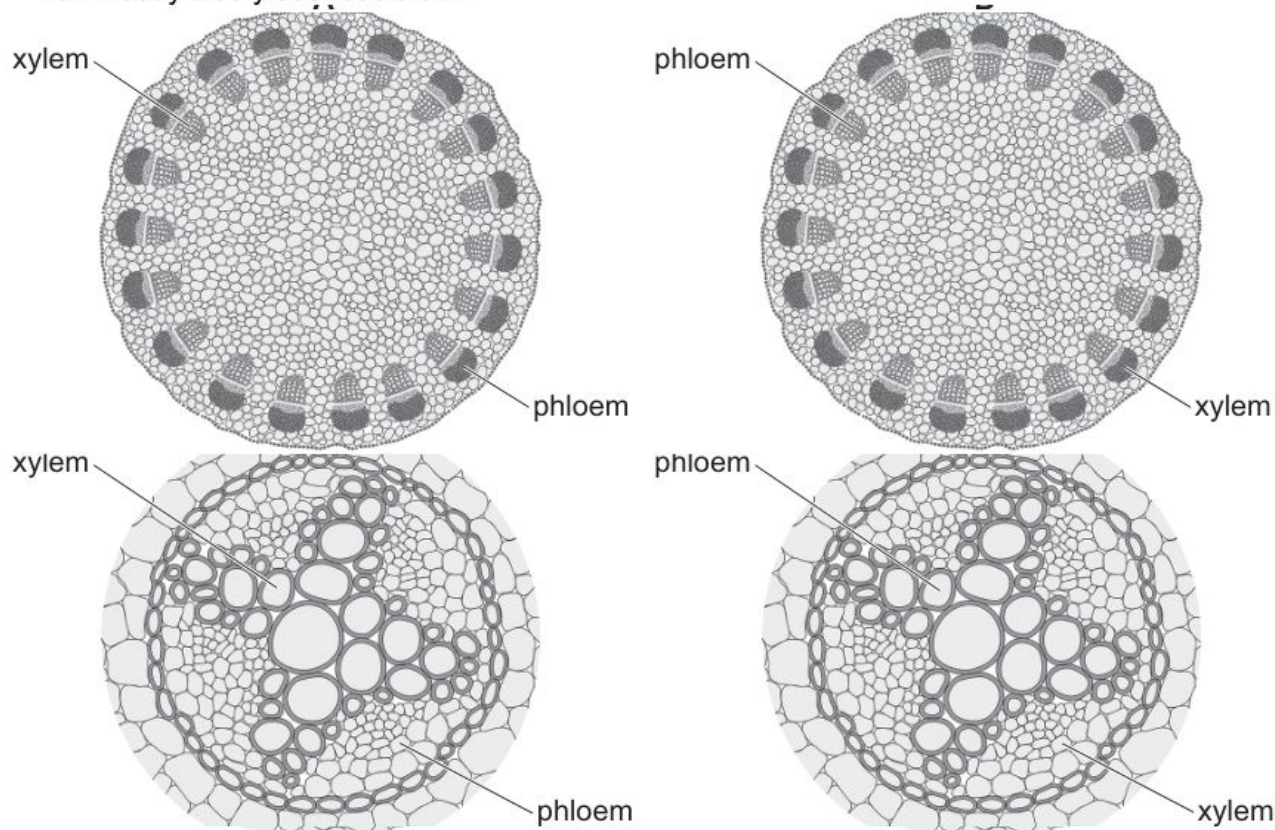
B

1

QUESTION

125 · 5090/11/M/J/25 Q12

12 Which transverse section photomicrograph shows the position of the xylem and phloem in a non-woody dicotyledonous stem?



13 Students set up apparatus to measure the uptake of water by a leafy shoot.

MARK SCHEME

5090/11/M/J/25 Q12

12

A

1

QUESTION

126 · 5090/11/M/J/25 Q23

23 Which structure is the ureter?



24 The processes of filtration and reabsorption take place in the kidneys.

MARK SCHEME

5090/11/M/J/25 Q23

23	C	1
----	---	---

QUESTION

127 · 5090/11/M/J/25 Q24

24 The processes of filtration and reabsorption take place in the kidneys. The daily amounts of some of the substances entering the kidneys in the blood and of these substances leaving the kidneys in the urine are shown.

substance	amount entering the kidneys in the blood	amount leaving the kidneys in the urine
water	150 dm ³	1.5 dm ³
glucose	150 g	0 g
urea	50 g	30 g

Which percentage of each substance is reabsorbed by the kidneys during a day?

	% of substance reabsorbed by kidneys		
	water	glucose	urea
B	148.5	150	20

MARK SCHEME

5090/11/M/J/25 Q24

24	A	1
----	---	---

QUESTION

128 · 5090/11/M/J/25 Q29

29 A nurse tests a sample of urine from a patient using Benedict's reagent. The sample tests what is a possible explanation for this result?

- A The patient's liver is **not** deaminating protein efficiently.
 B The patient's pancreas is **not** producing sufficient insulin.
 C The patient has eaten too much glucose.
 D The patient's concentration of plasma proteins is too high.

MARK SCHEME

5090/11/M/J/25 Q29

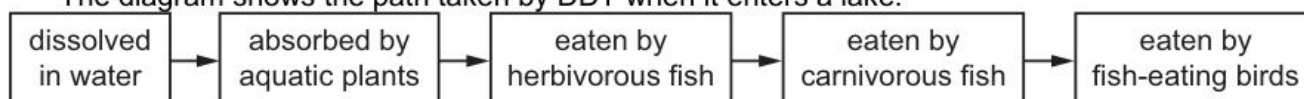
29	B	1
----	---	---

QUESTION

129 · 5090/11/M/J/25 Q40

40 DDT is an insecticide that is harmful to living organisms. It builds up in the cells of living organisms as it is **not** excreted.

The diagram shows the path taken by DDT when it enters a lake.



Which organisms would the DDT harm the most?

- A** aquatic plants
- B** carnivorous fish
- C** herbivorous fish
- D** fish-eating birds

MARK SCHEME

5090/11/M/J/25 Q40

40	D	1
----	---	---

QUESTION

130 · 5090/12/M/J/25 Q11

- 11 Glucose produced by photosynthesis can be converted into other carbohydrates or used immediately as glucose.

What is an immediate use of glucose in plants?

- A building cell walls
 B providing energy in respiration
 C storing energy
 D transporting energy in the phloem

MARK SCHEME

5090/12/M/J/25 Q11

11

B

1

QUESTION

131 · 5090/12/M/J/25 Q25

- 25 Which letter shows where urea is produced?



MARK SCHEME

5090/12/M/J/25 Q25

25

D

1

QUESTION

132 · 5090/12/M/J/25 Q26

- 26 Which substance would **not** be present in the urine of a healthy person with a normal diet?

B salts

- 27 Which part of the eye contains the greatest density of light receptors?

MARK SCHEME

5090/12/M/J/25 Q26

26

A

1

QUESTION

133 · 5090/11/O/N/25 Q5

- 5 Which process involves the movement of molecules against a concentration gradient?

- A active transport
 B diffusion
 D transpiration

MARK SCHEME

5090/11/O/N/25 Q5

5

A

1

QUESTION

134 · 5090/11/O/N/25 Q14

- 14 Which processes use energy obtained from respiration?
- 1 contraction of intercostal muscles during breathing
 - 2 nitrate ions entering a root hair cell against the concentration gradient
 - 3 oxygen moving from the lungs into blood capillaries
 - 4 production of antibodies by lymphocytes
 - 5 water entering a bacterial cell by osmosis
- A 1, 2 and 3 B 1, 2 and 4 C 2, 3 and 4 D 3, 4 and 5

MARK SCHEME

5090/11/O/N/25 Q14

14	B	1
----	---	---

QUESTION

135 · 5090/11/O/N/25 Q15

- 15 During vigorous exercise, lactic acid builds up in the muscles. Blood leaving these muscles will have a high concentration of lactic acid. Which process removes **most** of the lactic acid from the blood?
- A It is broken down in the liver.
 B It is excreted during sweating.
 C It is expired during faster breathing.
 D It is oxidised to release energy in muscles.
- 16 The table shows the mean rate of blood flow through some blood vessels.

MARK SCHEME

5090/11/O/N/25 Q15

15	A	1
----	---	---

QUESTION

136 · 5090/11/O/N/25 Q21

- 21 The table shows the concentration of five substances in the blood entering the kidney, in fluid entering the nephron and in the urine.

substance	concentration / mg per dm ³		
	blood entering the kidney	fluid entering the nephron	urine
urea	0.4	20.0	20.0
glucose	1.5	1.5	0.0
amino acids	0.8	0.8	0.0
ions	8.0	8.0	16.5
protein	82.0	0.0	0.0

Which substances are totally reabsorbed into the blood from the nephron?

- A amino acids and glucose
 B glucose and ions
 C protein and amino acids
 D protein and glucose
- 22 The graph shows blood glucose concentration over 6 hours in a day.

MARK SCHEME

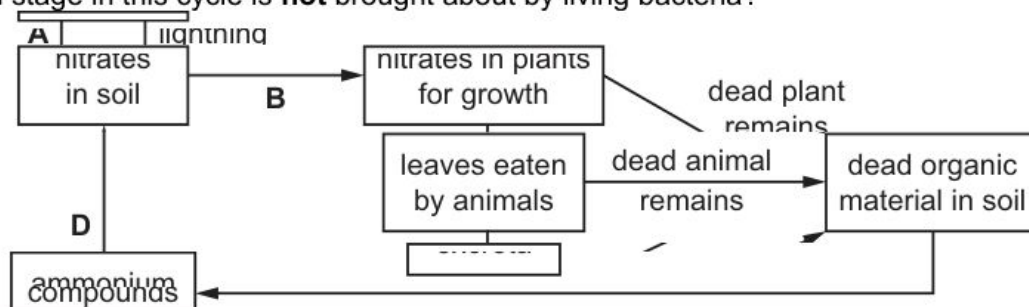
5090/11/O/N/25 Q21

21	A	1
----	---	---

QUESTION

137 · 5090/11/O/N/25 Q39

- 39 Trees in a forest need a constant supply of nitrates to continue growing. The diagram shows some of the stages of the nitrogen cycle that provide the necessary nitrates. Which stage in this cycle is **not** brought about by living bacteria?



MARK SCHEME

5090/11/O/N/25 Q39

39	B	1
----	---	---

QUESTION

138 · 5090/12/O/N/25 Q8

- 8 Oxygen is produced in leaf cells during photosynthesis and diffuses out of the leaf. By which pathway does the oxygen diffuse?

- A through the upper epidermis and through the cuticle
 B through the airspaces and through the stomata
 C through the mesophyll and through the phloem
 D through the mesophyll and through the xylem

MARK SCHEME

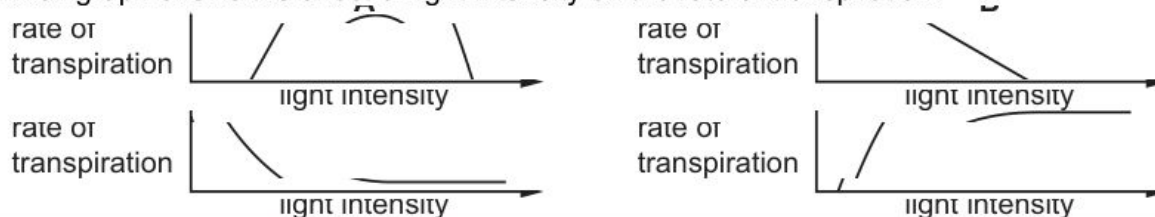
5090/12/O/N/25 Q8

8	B	1
---	---	---

QUESTION

139 · 5090/12/O/N/25 Q10

- 10 Which graph shows the effect of light intensity on the rate of transpiration?



MARK SCHEME

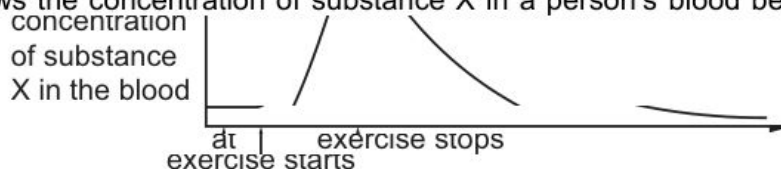
5090/12/O/N/25 Q10

10	D	1
----	---	---

QUESTION

140 · 5090/12/O/N/25 Q14

- 14 The graph shows the concentration of substance X in a person's blood before, during and after



What is substance X?

- C** lactic acid

MARK SCHEME

5090/12/O/N/25 Q14

14	C	1
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QUESTION

141 · 5090/12/O/N/25 Q23

- 23 Which term can be used to describe the release of oxygen from plants?

- B** excretion
C respiration
D transpiration

- 24 The diagram shows some structures in the skin.

MARK SCHEME

5090/12/O/N/25 Q23

23	B	1
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QUESTION

142 · 5090/12/O/N/25 Q35

- 35 Bread is made from a mixture of flour, sugar, water and yeast. The mixture is left in a warm place to allow it to rise before being baked. Which process makes the bread rise?

- A** The yeast feeds on the flour and grows.
B The yeast reproduces to make the loaf bigger.
C The yeast uses sugar to respire and produces carbon dioxide.
D The yeast produces alcohol which it excretes.

MARK SCHEME

5090/12/O/N/25 Q35

35	C	1
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